

salesforce

Language-based AI Agents and Large Action Models (LAMs)

CVPR 2024 Tutorial “Generalist Agent AI”

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He/Him



Conversational AI Assistants



Work Agent



Hello I'm Einstein
Copilot your trusted AI
assistant

Can you help route my
case for approval?



I have modified this
case status to Escalated
and created an
escalation request with
the subject "Modify
deliver center to
Portland, OR".

[Edit Case]



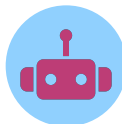
Personal Agent

Let's start booking my
summer vacation
travel. Find and book
family friendly hotel for
my upcoming stay in
France..



I found this nice hotel
near the venues you
are planning to visit. I
booked 2 rooms for the
full duration of your
stay.

[View Booking]



Creative Agent

I'm writing a fiction
story about animals in
the jungle. What are
some adventures
these animals might
have? Give me some
ideas and illustrate
them with pictures.



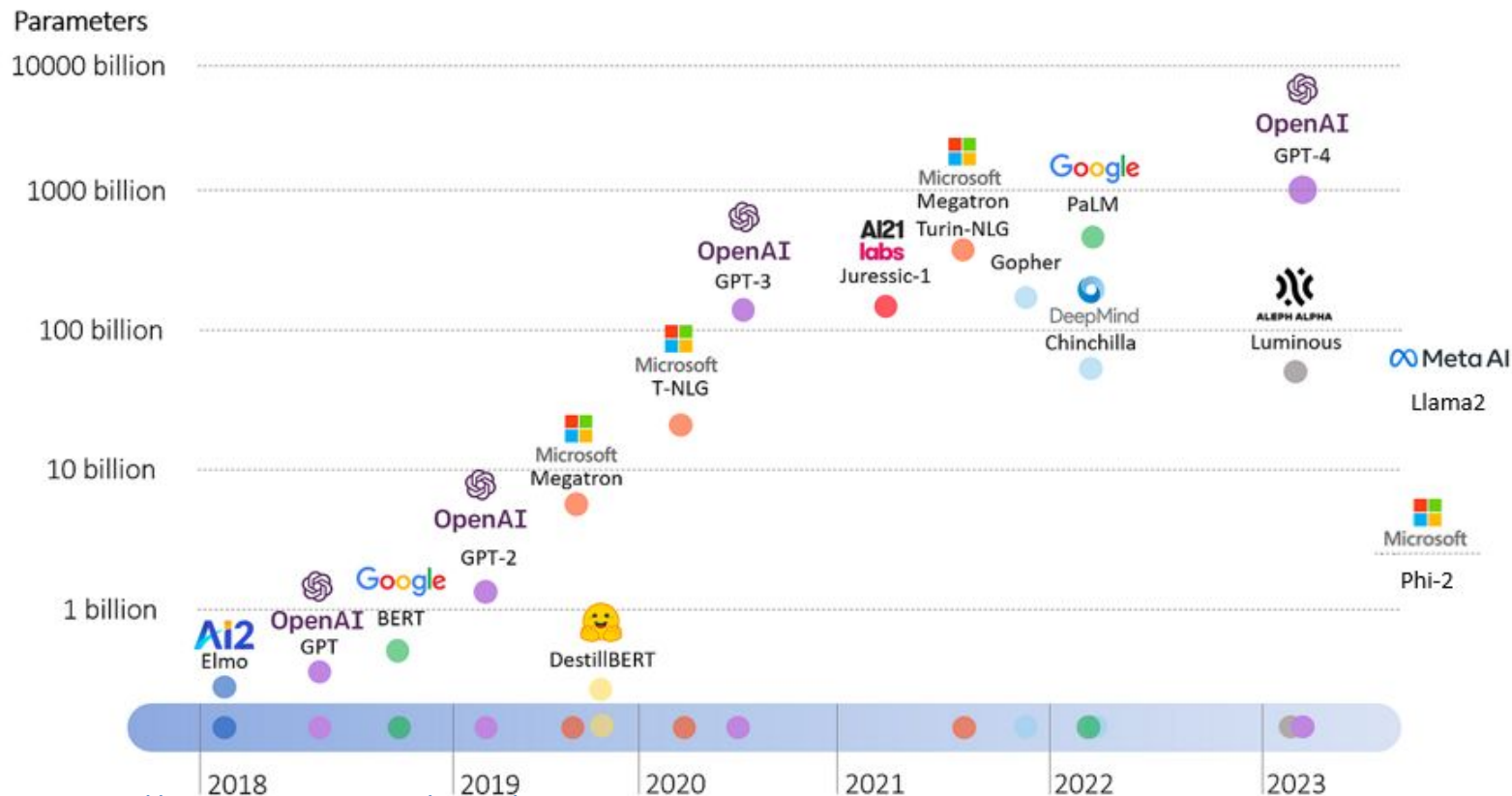
Sure, some ideas:
1. The animals might
organizing a party.



2. The animals may
need to solve a
problem [...]



LLM Parameter count



Source: <https://www.predli.com/post/llm-deep-dive-phi-2>

Language-based Agents



Research
Papers



open source

Huang et al, Zero-shot Planners, 2022

Yao et al, ReAct, 2022

Schick et al, Toolformer, 2023

Shen et al, HuggingGPT, 2023

Paul et al, Refiner, 2023

Shinn et al, Reflection, 2023

Madaan et al, Self-refine, 2023

Park et al, Generative Agents, 2023

...



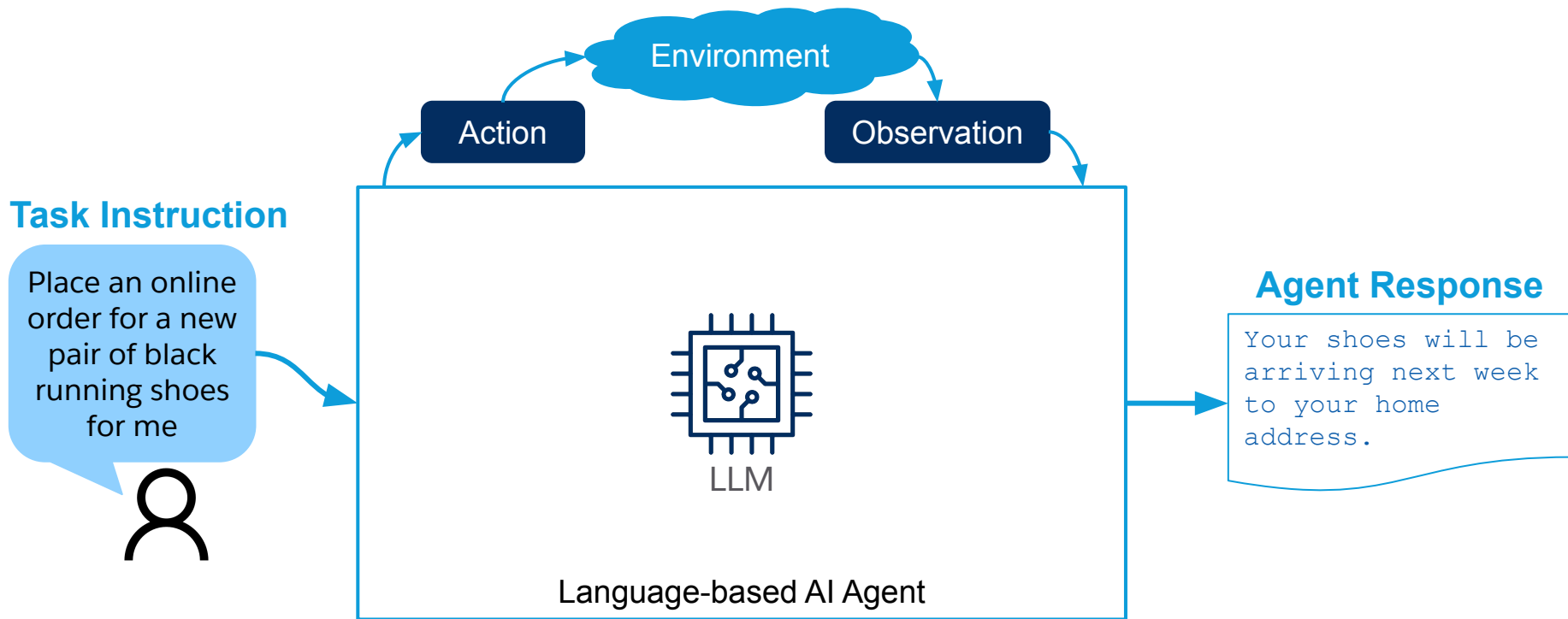
LangChain



BabyAGI

...

High-level Language-based Agent framework



Agent Benchmarking

What tasks are agents solving today?



[illegible]

- "User_query":**

"steps":

"next observation": "OK."

"next observation": "Arthur's Magazine was published in 1844."

"next observation": "First for Women is a woman's magazine published in 1989."

"content": "Action: Finish[Arthur's Magazine]",

"next observation": "Answer is CORRECT."

[Yang et al. "HotPotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering". EMNLP 2018]



WebShop - Decision Making

<https://webshop-pnlp.github.io/>



User gives natural language instruction

The image shows a screenshot of the WebShop interface. At the top, there is a shopping cart icon and the text "WebShop". Below this, a text box contains the instruction: "i am looking for x-large, red color women faux fur lined winter warm jacket coat, and price lower than 70.00 dollars". This text box is highlighted with an orange oval. Below the instruction box is a search bar with the placeholder text "Search..." and a green "Search" button. At the bottom, there are two blue buttons: "Project Site" and "Task Instructions".

- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal



WebShop - Decision Making

<https://webshop-pnlp.github.io/>



- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal

Current Action: `search[red faux fur coat winter jacket]`



WebShop - Decision Making



<https://webshop-pnlp.github.io/>

Instruction: i am looking for x-large, red color women faux fur lined winter warm jacket coat, and price lower than 70.00 dollars

Current Query: red faux fur coat winter jacket

Results

Page 1 (1-10) of 50 total results

[Back to Search](#)

[Next >](#)



B09KP78G37

Women Faux Fur Lined Jacket Coat
Winter Warm Thick Fleece Outwear
Trench Zipper Plus Size Long
Slava Blue Outwear



B07ZXBGDXF

Women's Coat, FORUU Winter Faux
Fur Fleece Outwear Warm Lapel
Biker Motor Aviator Jacket



B000J3HZXS

Laguna Floating Planting Bag,
Aquatic Pond Planter

★★★★ 4.4

Current Action: click[Women's Coat, FORUU Winter...]

- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal



WebShop - Decision Making



<https://webshop-pnlp.github.io/>

Instruction: i am looking for x-large, red color women faux fur lined winter warm jacket coat, and price lower than 70.00 dollars
Current Query: red faux fur coat winter jacket

Back to Search

< Back to Results



Women's Coat, FORUU Winter Faux Fur Fleece Outwear Warm Lapel Biker Motor Aviator Jacket

Buy Now

Price: \$21.49 to \$24.99

Rating: 2.4

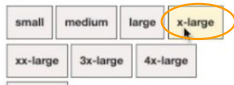
Description

Features

Reviews

Attributes

size



Current Action: click [x-large]



- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal



WebShop - Decision Making



<https://webshop-pnlp.github.io/>

Instruction: i am looking for x-large, red color women faux fur lined winter warm jacket coat, and price lower than 70.00 dollars
Current Query: red faux fur coat winter jacket

Back to Search

< Back to Results



Women's Coat, FORUU Winter Faux Fur Fleece Outwear Warm Lapel Biker Motor Aviator Jacket

Price: \$21.49 to \$24.99

Rating: 2.4

Buy Now

Description

Features

Reviews

Attributes

size



Current Action: click[red]



- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal



WebShop - Decision Making



<https://webshop-pnlp.github.io/>

Instruction: i am looking for x-large, red color women faux fur lined winter warm jacket coat, and price lower than 70.00 dollars

Current Query: red faux fur coat winter jacket

[Back to Search](#) [< Back to Results](#)



Women's Coat, FORUU Winter Faux Fur Fleece Outwear Warm Lapel Biker Motor Aviator Jacket

Price: \$21.49 to \$24.99

Rating: 2.4

[Description](#) [Features](#) [Reviews](#) [Attributes](#)

[Buy Now](#)

size

small	medium	large	x-large
xx-large	3x-large	4x-large	

Current Action: click[buy now]

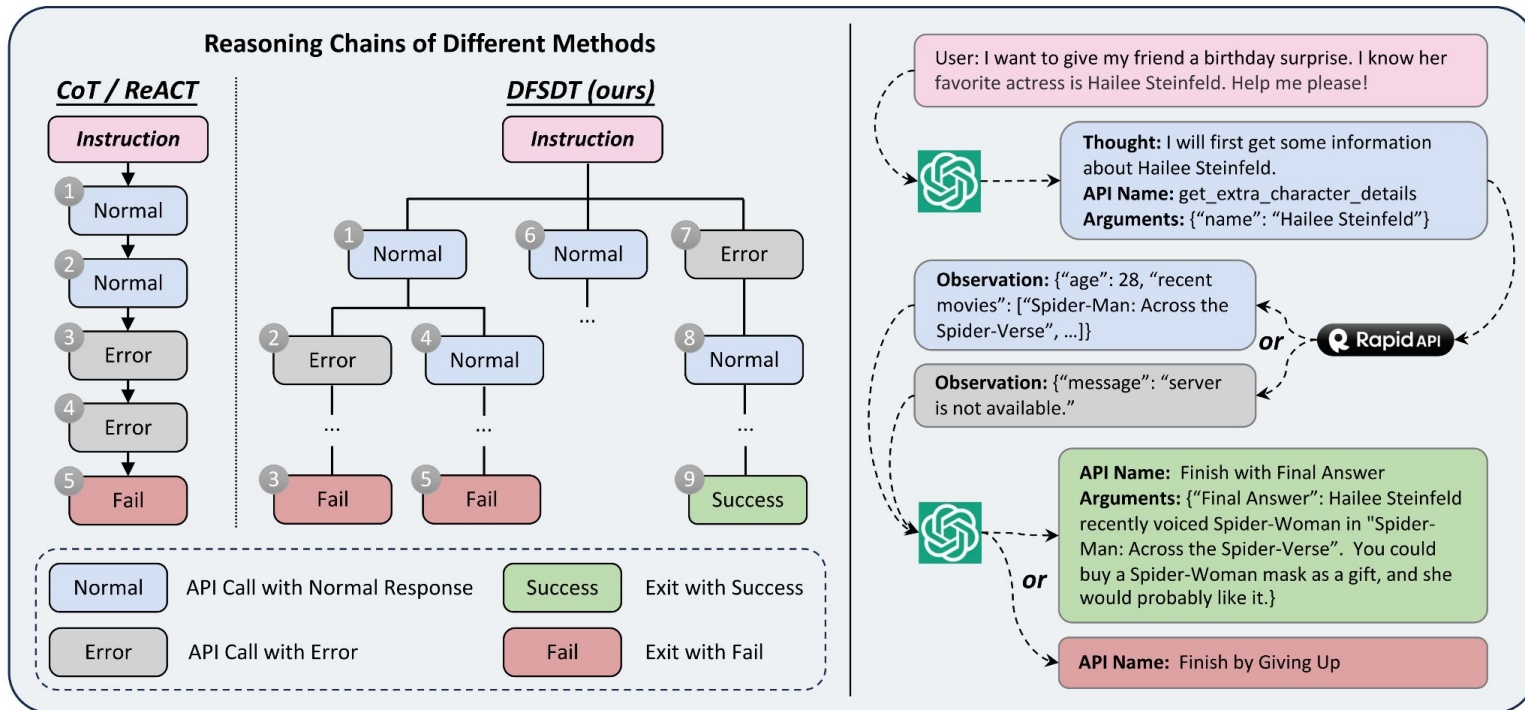
gray	red	black
------	-----	-------

- Simplified e-commerce shopping environment containing **1.18M real-world products**
- **12k human instructions, ground truth items**
- Agent receives single instruction from user, then performs multiple actions to achieve the goal
- Interaction is purely text-based



ToolBench – Tool-use capability

<https://github.com/OpenBMB/ToolBench>





ToolBench – Tool-use capability

<https://github.com/OpenBMB/ToolBench>

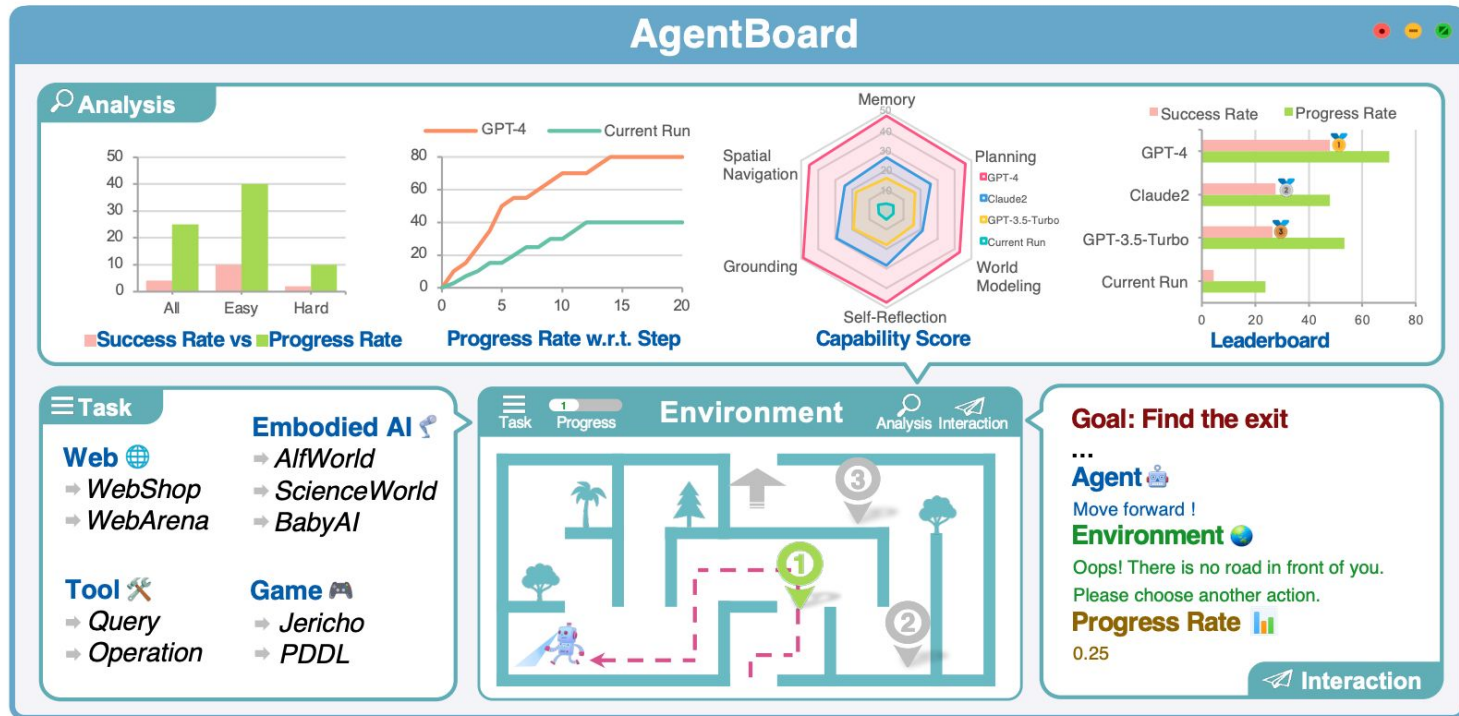


Resources	ToolBench
Number of tools	3,451
Number of APIs	16,461
Avg. Reasoning Traces	4
Number of evaluation queries	1,100
Evaluation metrics	- Pass Rate - Win Rate



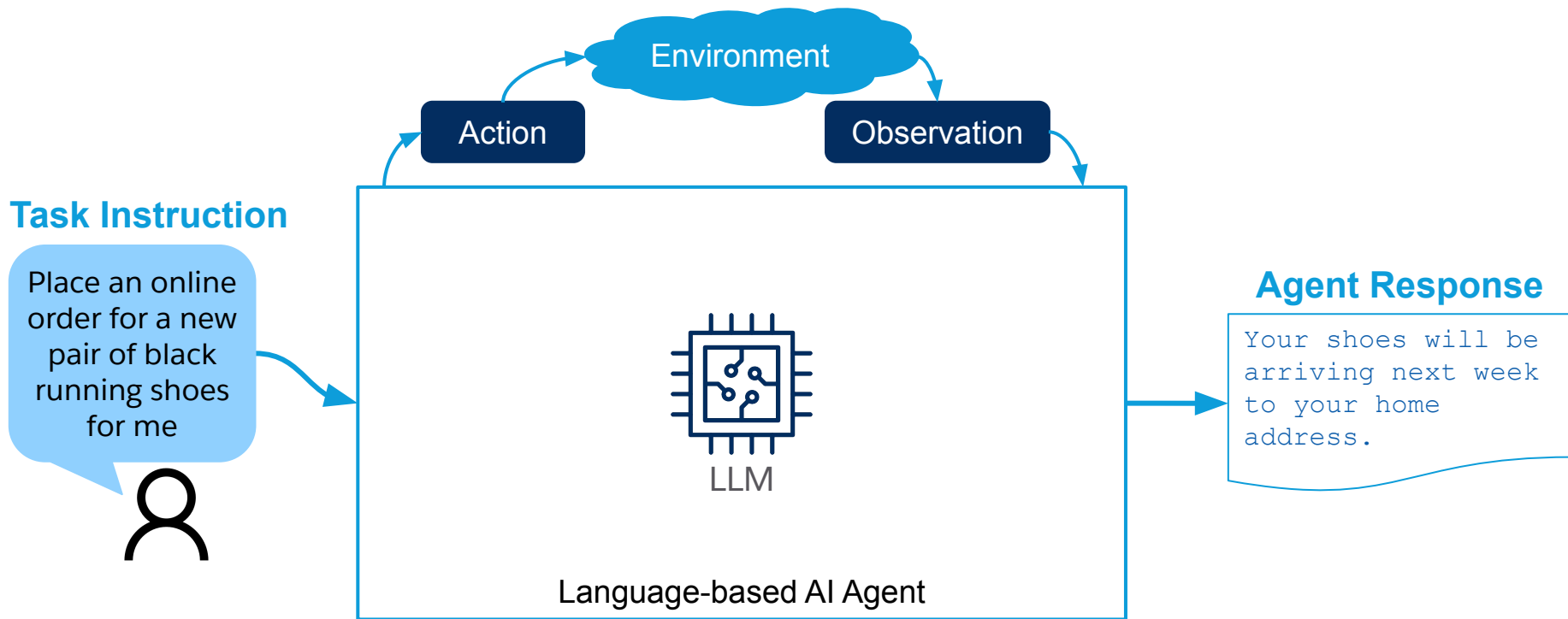
AgentBoard: evaluation framework

<https://github.com/hkust-nlp/AgentBoard>



[Ma et al "AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents", 2023]

High-level Language-based Agent framework

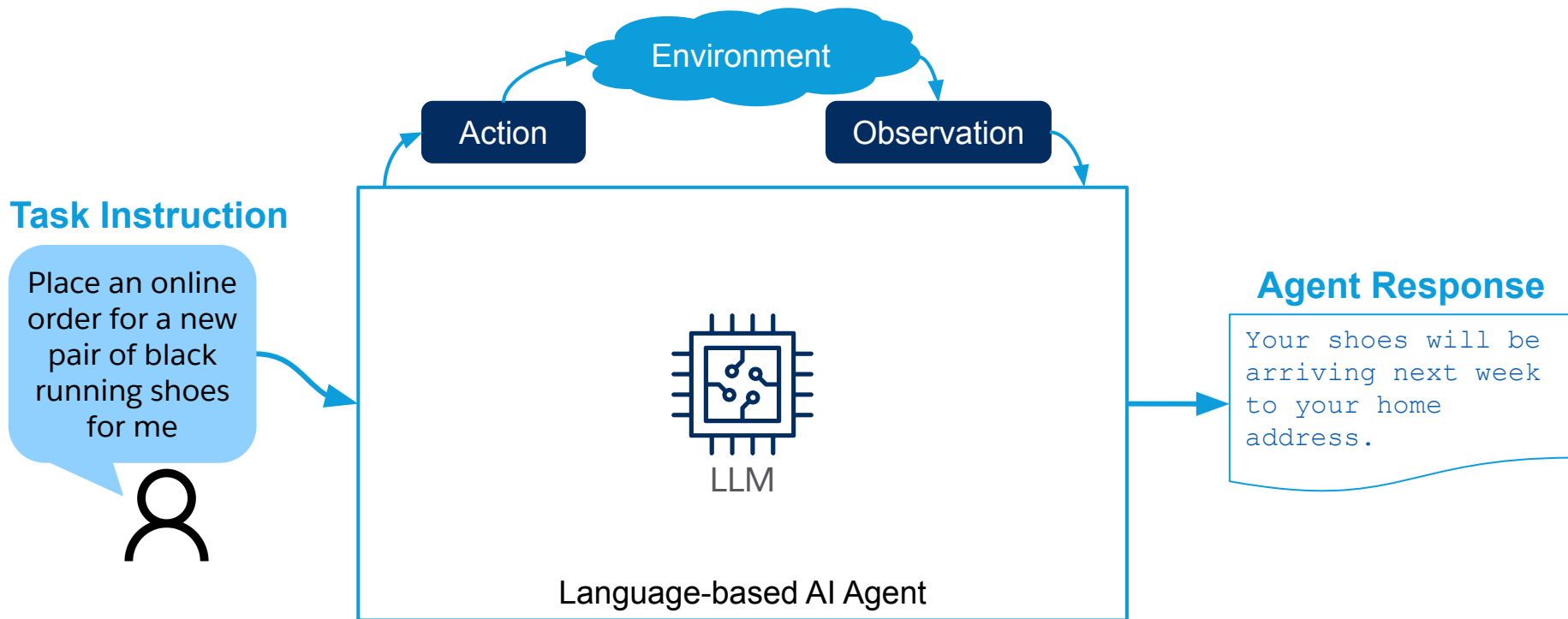


Agent Architectures

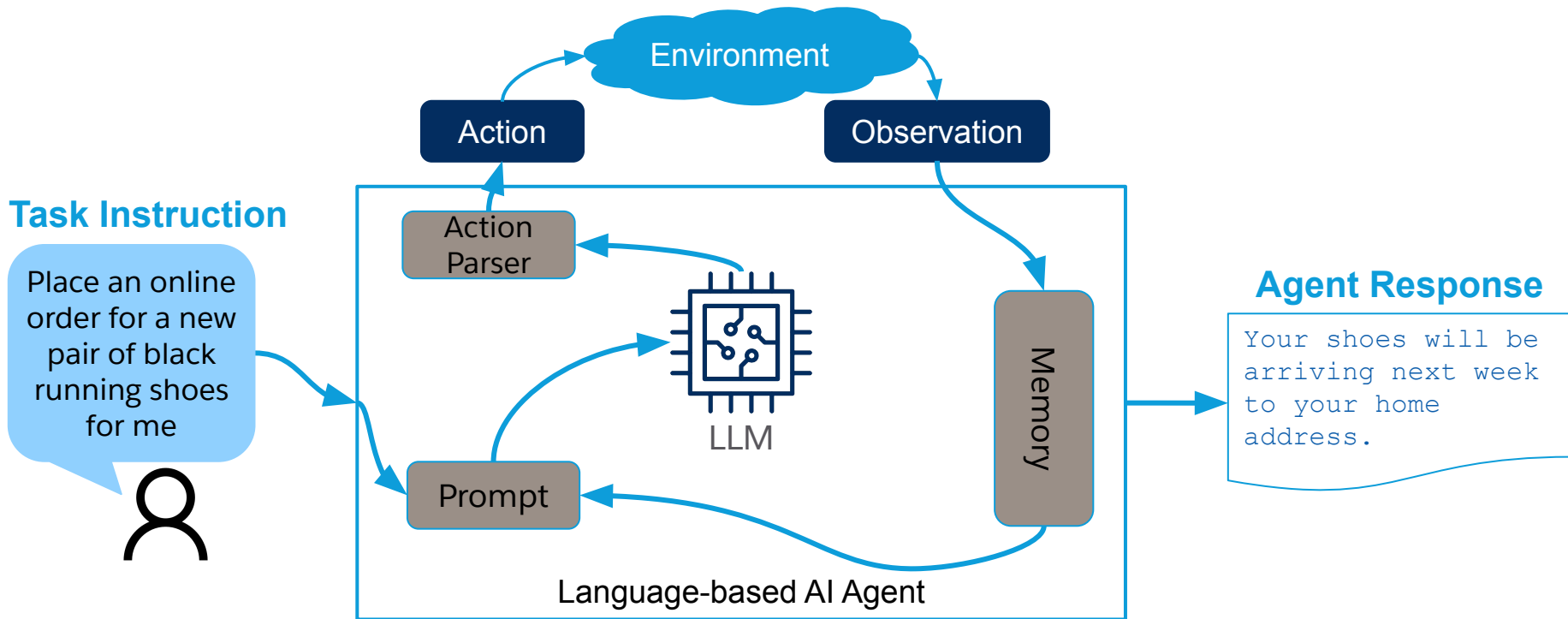
How are we building Agents today?



High-level Language-based Agent framework



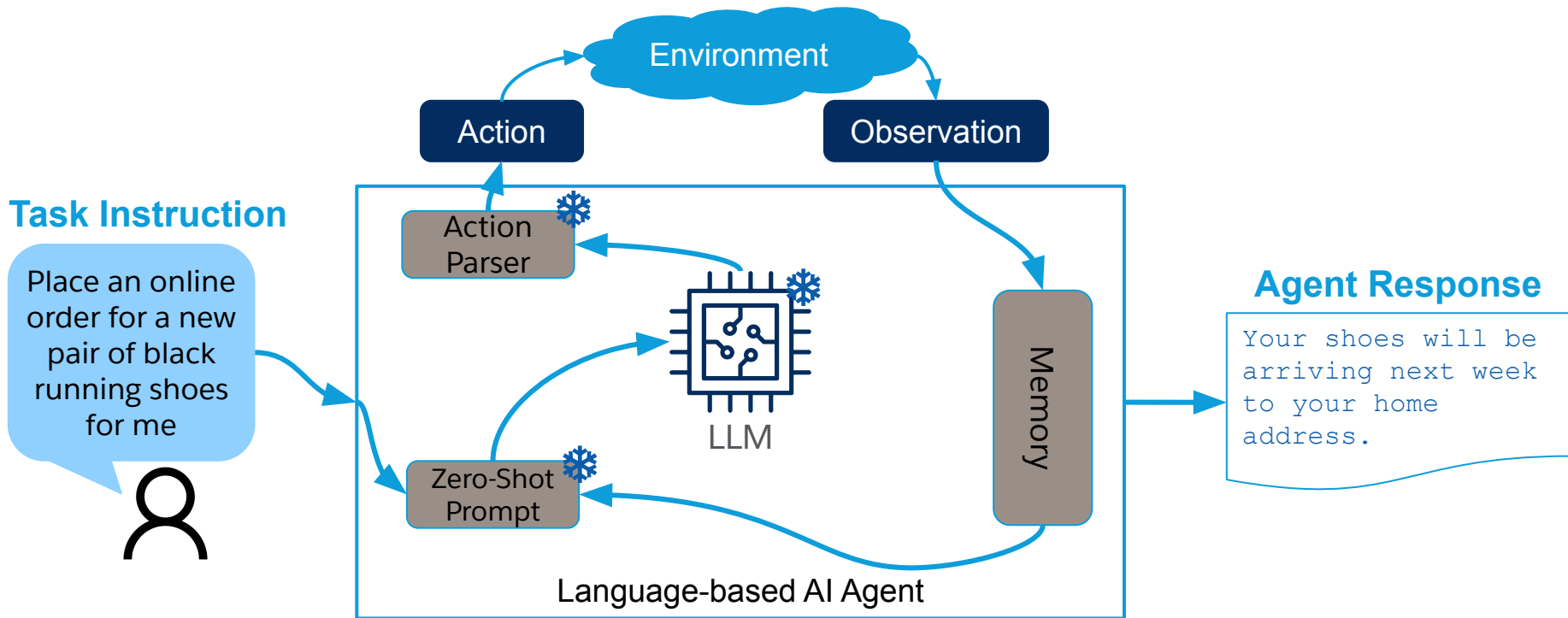
High-level Language-based Agent framework



Language-based AI Agents



Zero-shot Language-based Agent



Zero-shot Agent Example



System Prompt

You are an intelligent agent. You should follow your [Role], [Action_Doc] to take actions. Your generation should follow the example format. Finish the task as best as you can.

[Role] You can interact with the webshop. [End of Role]

[Constraint] Your generation should be simple and clear. [End of Constraint]

[Action_Doc] [{ 'name': 'search', 'description': '<fc_description>', 'parameters': { 'product': '<param_description>' } }, { 'name': 'click', 'description': '<fc_description>', 'parameters': { 'button': '<param_description>' } }, { 'name': 'Finish', 'description': '<fc_description>', 'parameters': { 'response': '<param_description>' } }] [End of Action_Doc]

Using the following action format example to generate well formatted actions.

[ActionFormatExample]

Action:search[{"product": "the name of the product to search for"}]

Action:Finish[{"response": "this is the finish action response. Respond towards the task instruction."}]

Action:click[{"button": "the name of the button to click"}]

[End of ActionFormatExample]

[Zhiwei Liu. "BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents". ICLR 2024 Workshop on LLM Agents]

Zero-shot Agent Example



Task Instruction: I need a long clip-in hair extension which is natural looking, and price lower than 40.00 dollars

[Execution]

Action: search[{"product": "long clip-in hair extension"}]

Action: click[{"button": "B09C337K8S"}]

Action:

Finish[{"response": "I found a long clip-in hair extension that is natural looking and priced lower than \$40.00. It is the Hair Clip Extensions for Women Natural Looking Hair Extensions Clip in Hairpieces Black Synthetic Hair Extensions, priced at \$29.99."}]

[Environment]

Observation: WEB PAGE: {
[Back to Search] Page 1 (Total results: 15)
[Next >]
[B09C337K8S]
Hair Clip Extensions for Women Natural Looking Hair
Extensions clip in Hairpieces Black Synthetic Hair Extensions
\$29.99
[B099K9Z9L2]
Clip in Hair Extensions 20" Double Weft 100% Remy
Human Hair Grade 8A Quality 7pcs 16clips 70grams \$43.99
}

Observation: WEB PAGE: {
[Back to Search]
[< Prev]
Hair Clip Extensions for Women Natural Looking Hair Extensions Clip in
Hairpieces Black Synthetic Hair Extensions Price: \$29.99
Rating: N.A.
[Buy Now]
}

[Zhiwei Liu. "BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents". ICLR 2024 Workshop on LLM Agents]

Zero-shot Agent WebShop Results: Average Reward



LLM	Length	Zero-Shot
fastchat-t5-3b	2k	0.40
vicuna-7b	2k	0.00
vicuna-13b	2k	0.03
vicuna-33b	2k	0.14
llama-2-7b	4k	0.00
llama-2-13b	4k	0.07
llama-2-70b	4k	0.01
mpt-7b-instruct	8k	0.00
mpt-30b-instruct	8k	0.17
xgen-8k-7b-instruct	8k	0.00
longchat-7b-16k	16k	0.02
longchat-13b-16k	16k	0.00
text-davinci-003	4k	0.53
gpt-3.5-turbo	4k	0.51
gpt-3.5-turbo-16k	16k	0.57

- Bigger LLMs tend to achieve better agent performance out of the box
- Some LLMs struggle in Agent tasks in the zero-shot setting
- Big gap open source <-> GPT models

[Zhiwei Liu. "BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents". ICLR 2024 Workshop on LLM Agents]

Zero-shot Agent HotPotQA Results

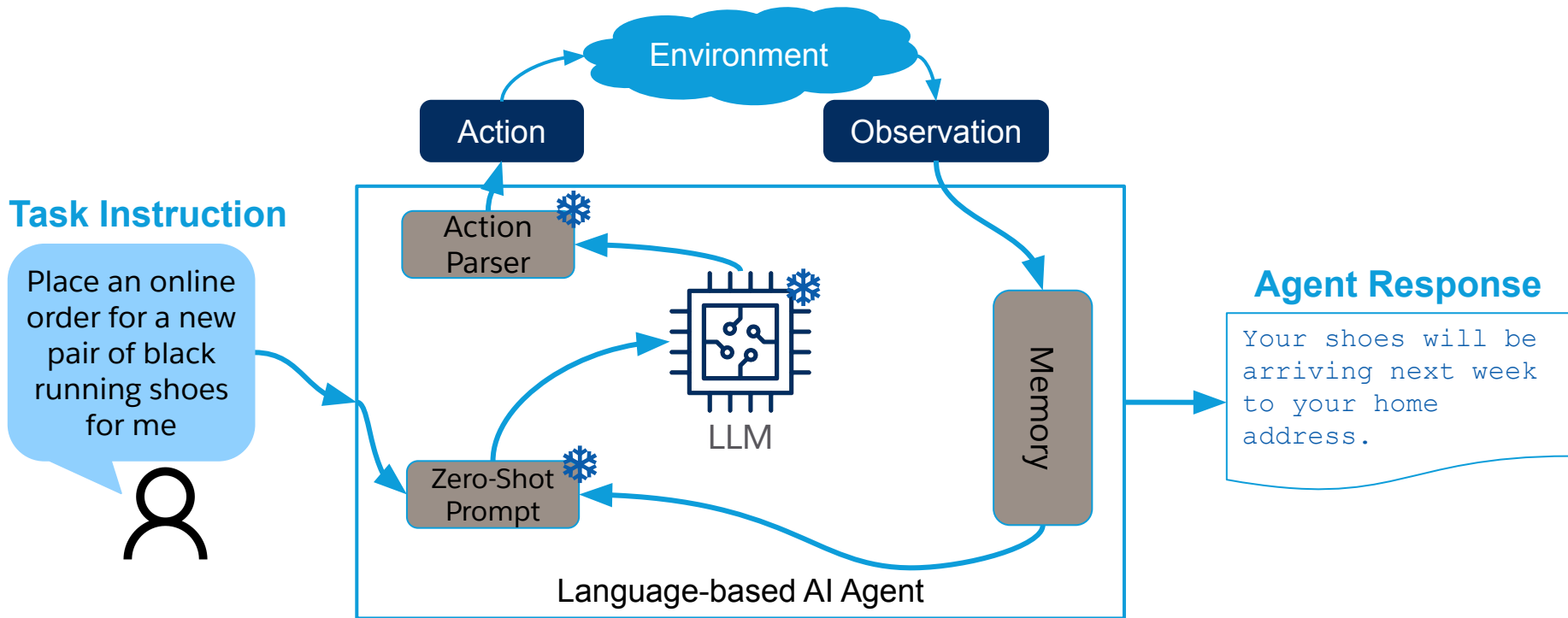


LLM	Length	Zero-Shot
fastchat-t5-3b	2k	0.03
vicuna-7b	2k	0.13
vicuna-13b	2k	0.15
vicuna-33b	2k	0.22
llama-2-7b	4k	0.04
llama-2-13b	4k	0.17
llama-2-70b	4k	0.28
mpt-7b-instruct	8k	0.10
mpt-30b-instruct	8k	0.16
xgen-8k-7b-instruct	8k	0.15
longchat-7b-16k	16k	0.08
longchat-13b-16k	16k	0.11
text-davinci-003	4k	0.34
gpt-3.5-turbo	4k	0.33
gpt-3.5-turbo-16k-0613	16k	0.30

- GPT models ahead again

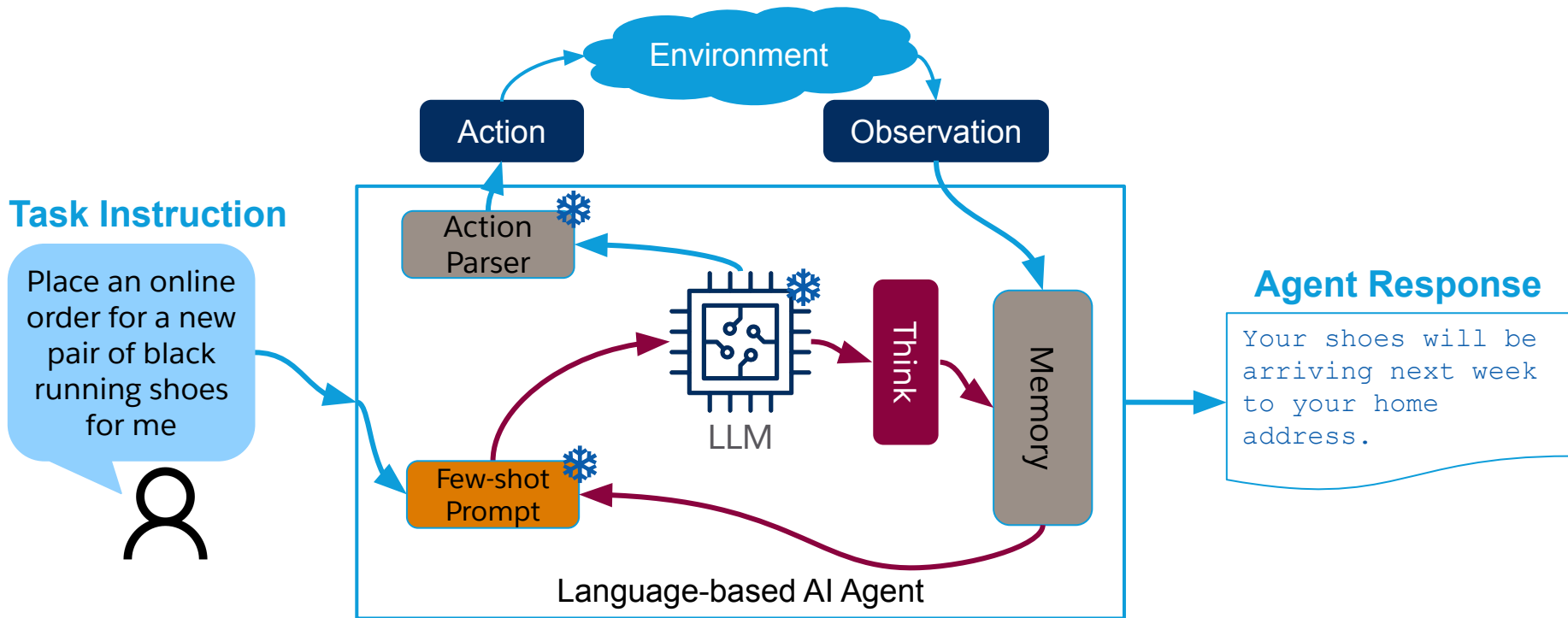
[Zhiwei Liu. “BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents”. ICLR 2024 Workshop on LLM Agents]

Zero-shot Language-based Agent



ReAct Agent

<https://github.com/ysymyth/ReAct>



ReAct Agent Example



System Prompt

You are an intelligent agent. You should follow your [Role], [Action_Doc] to take actions. Your generation should follow the example format. Finish the task as best as you can.

[Role] You can interact with the webshop. [End of Role]

[Constraint] You generation should be simple and clear. [End of Constraint]

[Action_Doc] [{'name': 'search', 'description': 'search for a product in the webshop', 'parameters': {'product': 'the name of the product to search for'}}, {'name': 'Finish', 'description': 'Complete the task with a response.', 'parameters': {'response': 'this is the finish action response. Respond towards the task instruction.'}}, {'name': 'click', 'description': 'click a button in the web page', 'parameters': {'button': 'the name of the button to click'}}, {'name': 'Think', 'description': 'Conduct thinking and reasoning process for solving task.', 'parameters': {'response': 'this is your thinking response. Be specific and critical.'}}] [End of Action_Doc]

Using the following action format example to generate well formatted actions.

[ActionFormatExample]

Action:search[{"product": "the name of the product to search for"}]

Action:Finish[{"response": "this is the finish action response. Respond towards the task instruction."}]

Action:click[{"button": "the name of the button to click"}]

[End of ActionFormatExample]

ReAct Agent Example



[Example]

Task: i would like a 3 ounce bottle of bright citrus deodorant for sensitive skin, and price lower than 50.00 dollars

Action: Think[{"response": "I should first use Search to search bright citrus deodorant for sensitive skin."}]

Observation: OK

Action: search[{"product": "bright citrus deodorant for sensitive skin"}]

Observation: WEB PAGE: {

[Back to Search]

[< Prev]

scent [assorted scents][bright citrus][calming lavender] size [3 ounce (pack of 1)]}

Action: click[{"button": "bright citrus"}]

Observation: "You have clicked bright citrus."

Action: click[{"button": "3 ounce (pack of 1)"}]

Observation: "You have clicked 3 ounce (pack of 1)."

Action: click[{"button": "Buy Now"}]

ReAct Agent Example



Task Instruction: I need a long clip-in hair extension which is natural looking, and price lower than 40.00 dollars

Action: Think[{"response": "I should first use Search to search for a long clip-in hair extension."}]

Observation: OK

Action: search[{"product": "long clip-in hair extension"}]

Observation: WEB PAGE:

[Back to Search] Page 1 (Total results: 15) [Next >]
[B09C337K8S]

Hair Clip Extensions for Women Natural Looking
Hair Extensions Clip in Hairpieces Black Synthetic Hair
Extensions \$29.99

[B093BKWHFK]

MJMOJO Hair Extensions Clip in Human Hair
Extensions 120g 7pcs Straight Natural Black Hair Clip in
(16 inches #NST) \$63.99

Action: Think[{"response": "The first product is most relevant to the requirements. I should click the first product."}]

Observation: OK

ReAct Agent Example (continued)



Action: click[{"button": "B09C337K8S"}]

Observation: WEB PAGE:
[Back to Search] [< Prev]
Hair Clip Extensions for Women
Natural Looking Hair Extensions Clip in
Hairpieces Black Synthetic Hair
Extensions Price: \$29.99 Rating: N.A.
[Buy Now] }

Action: click[{"button": "Buy Now"}]

Observation: shopping is finished.

Action: Finish[{"response": "Task Finished."}]

Observation: Task Finished. Reward: 1.0

ReAct Agent WebShop Results



LLM	Length	Zero-Shot	ReAct
fastchat-t5-3b	2k	0.40	0.31
vicuna-7b	2k	0.00	0.10
vicuna-13b	2k	0.03	0.15
vicuna-33b	2k	0.14	0.19
llama-2-7b	4k	0.00	0.12
llama-2-13b	4k	0.07	0.26
llama-2-70b	4k	0.01	0.44
mpt-7b-instruct	8k	0.00	0.06
mpt-30b-instruct	8k	0.17	0.31
xgen-8k-7b-instruct	8k	0.00	0.07
longchat-7b-16k	16k	0.02	0.07
longchat-13b-16k	16k	0.00	0.24
text-davinci-003	4k	0.53	0.55
gpt-3.5-turbo	4k	0.51	0.54
gpt-3.5-turbo-16k	16k	0.57	0.49

- ReAct helps smaller models boost their performance.
- Big models/GPTs do not benefit much.

ReAct Agent HotPotQA Results

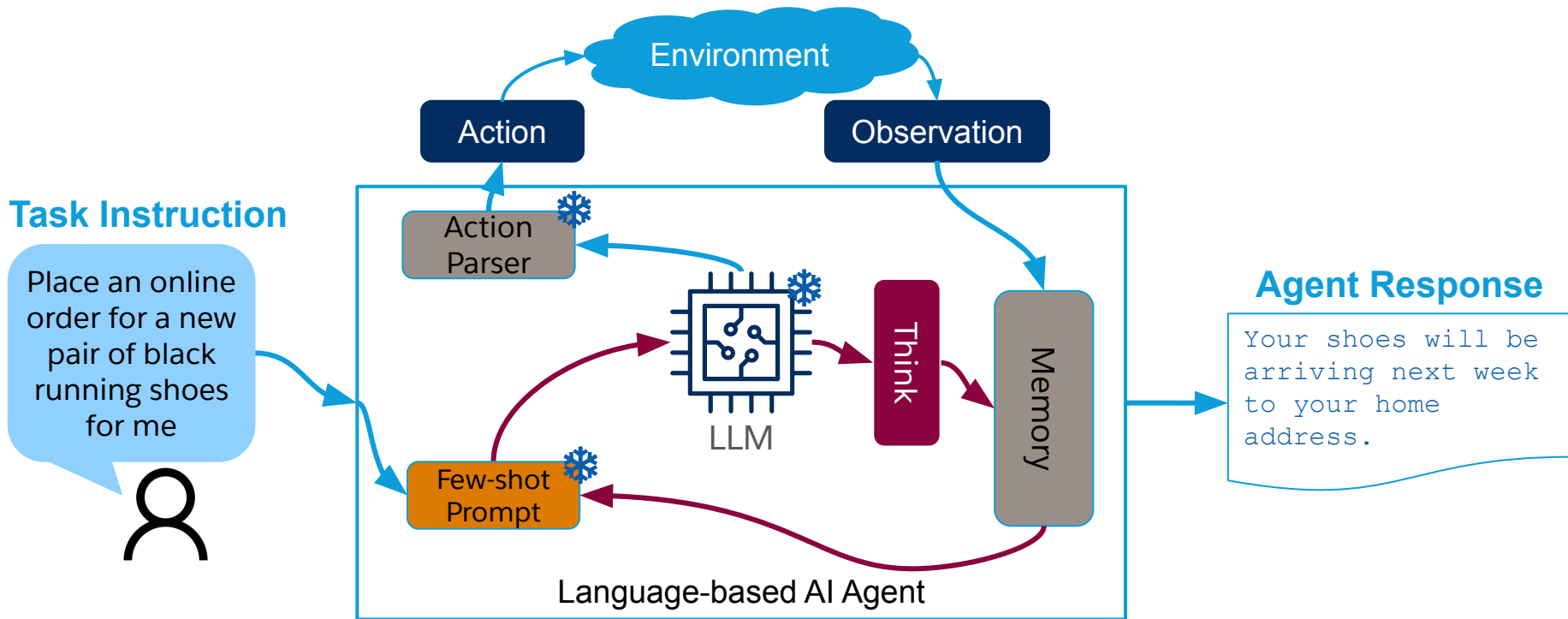


LLM	Length	Zero-Shot	ReAct
fastchat-t5-3b	2k	0.03	0.07
vicuna-7b	2k	0.13	0.03
vicuna-13b	2k	0.15	0.26
vicuna-33b	2k	0.22	0.26
llama-2-7b	4k	0.04	0.26
llama-2-13b	4k	0.17	0.25
llama-2-70b	4k	0.28	0.36
mpt-7b-instruct	8k	0.10	0.17
mpt-30b-instruct	8k	0.16	0.33
xgen-8k-7b-instruct	8k	0.15	0.19
longchat-7b-16k	16k	0.08	0.22
longchat-13b-16k	16k	0.11	0.24
text-davinci-003	4k	0.34	0.45
gpt-3.5-turbo	4k	0.33	0.32
gpt-3.5-turbo-16k-0613	16k	0.30	0.19

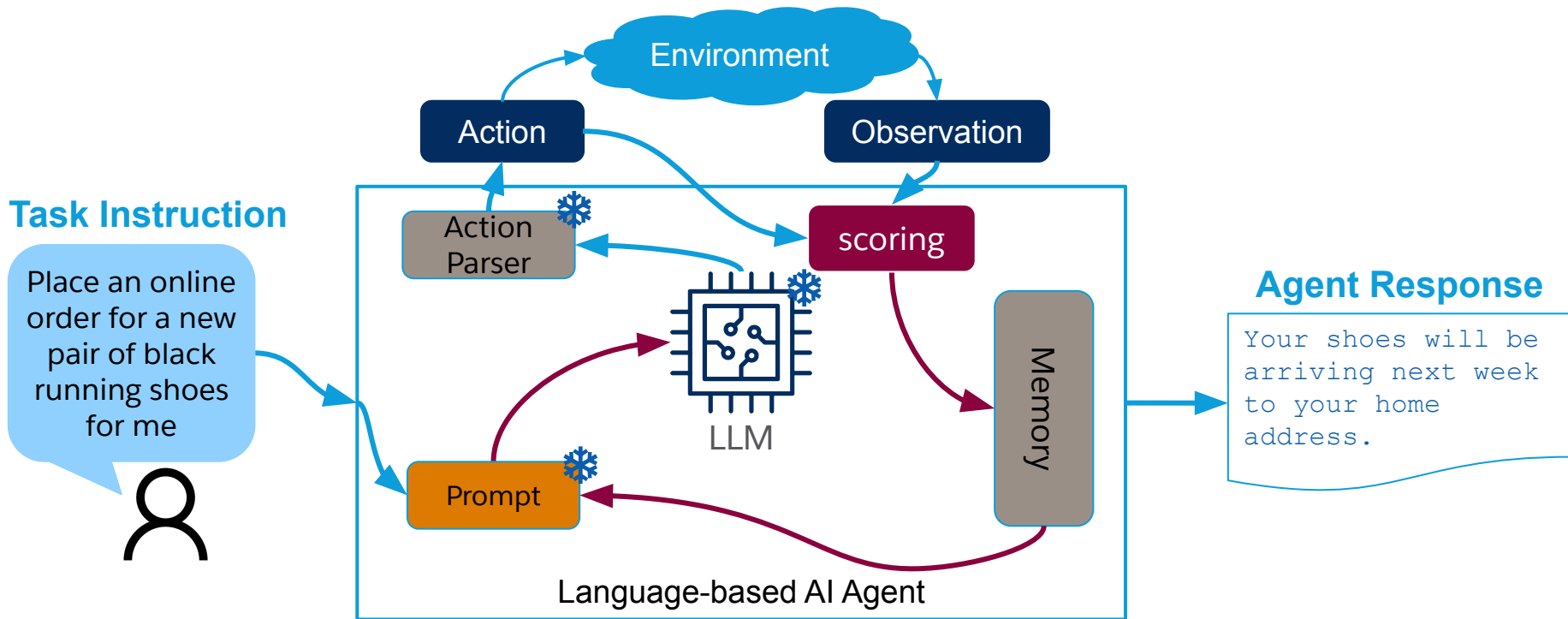
- ReAct helps smaller models boost their performance.
- Big models/GPTs do not benefit much.
- Model size more important than context length

ReAct Agent

<https://github.com/ysymyth/ReAct>



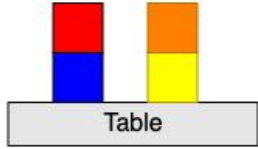
REX Agent Architecture



[Murthy, et al. "REX: Rapid Exploration and eXploitation for AI agents". ICLR 2024 LLM Agents Workshop.]

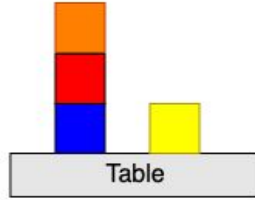
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free

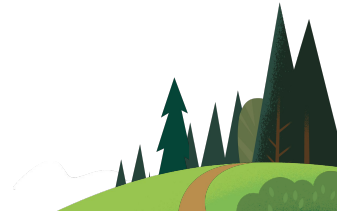


AI Agent

Pass-1

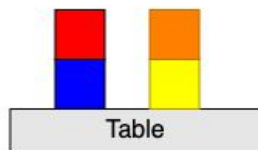


REX: UCB-CoT



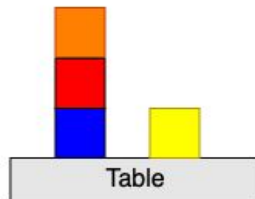
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



Pass-1 Agent's Output:

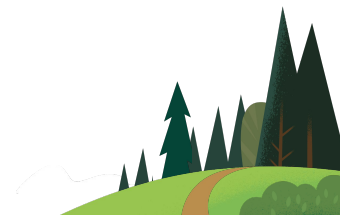
Step 1: Unstack Orange block from on top of Yellow block

Step 2: Pick Red block

Pass-1

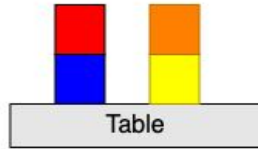


REX: UCB-CoT



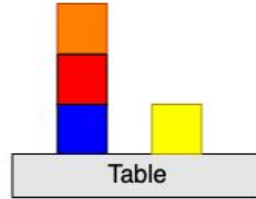
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Pick Red block

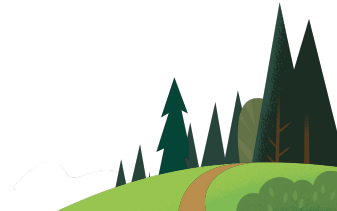
Step 1

Step 2

Pass-1

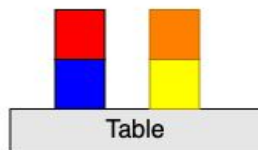
salesforce

REX: UCB-CoT



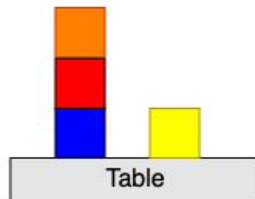
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



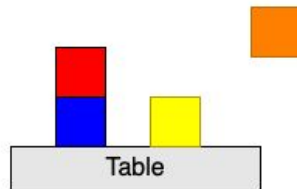
AI Agent



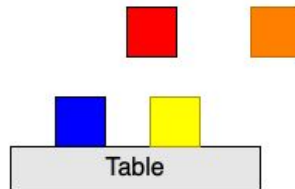
Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Pick Red block



Step 1

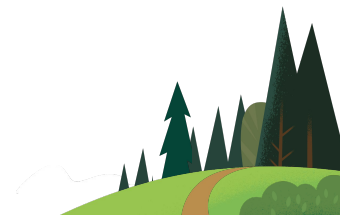


Step 2

Pass-1

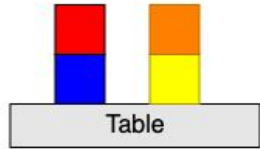


REX: UCB-CoT



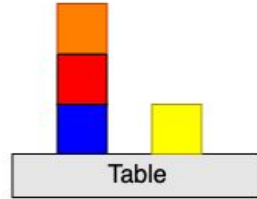
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



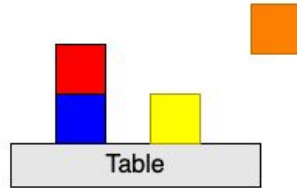
AI Agent



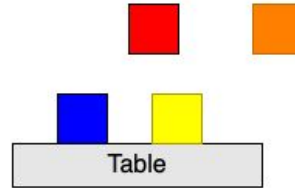
Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Pick Red block



Step 1



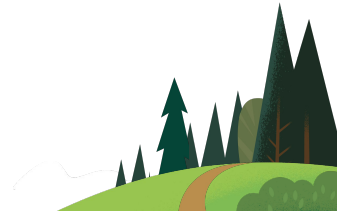
Step 2

Pass-1

salesforce

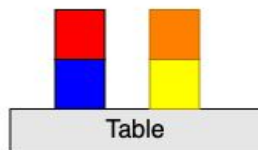
REX: UCB-CoT

Final Config



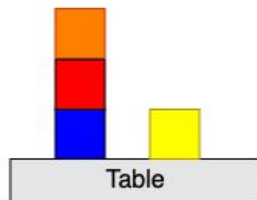
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



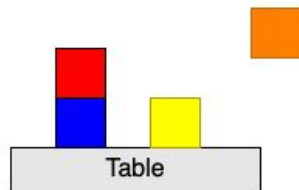
AI Agent



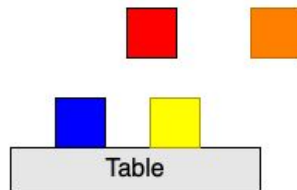
Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

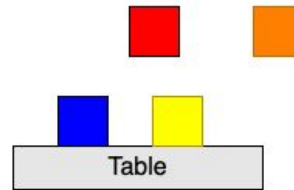
Step 2: Pick Red block



Step 1



Step 2

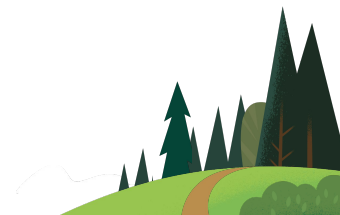


Final Config

Pass-1

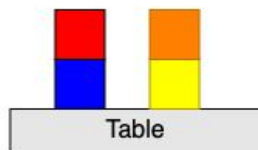


REX: UCB-CoT



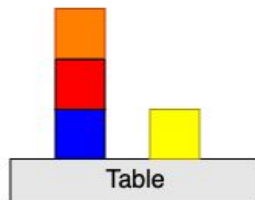
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



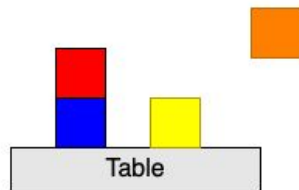
AI Agent



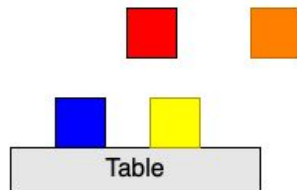
Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

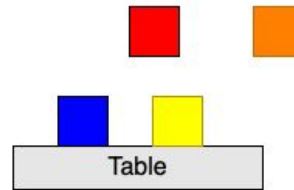
Step 2: Pick Red block



Step 1



Step 2



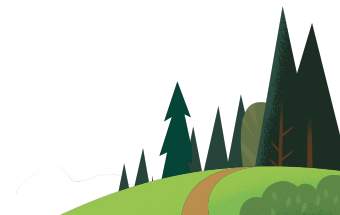
Final Config

Correct Answer?

Pass-1

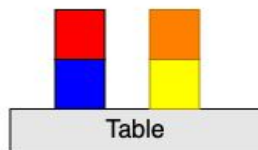
salesforce

REX: UCB-CoT



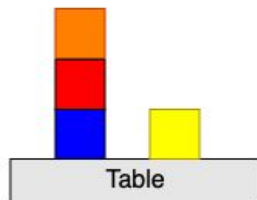
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



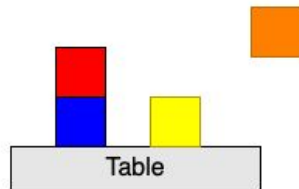
AI Agent



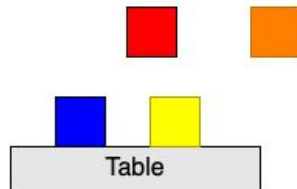
Pass-1 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

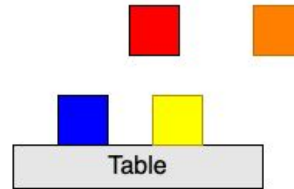
Step 2: Pick Red block



Step 1



Step 2



Final Config

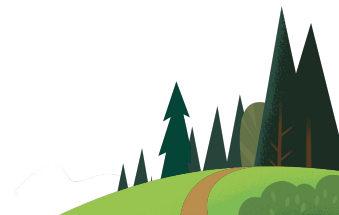
Correct Answer?



Pass-1

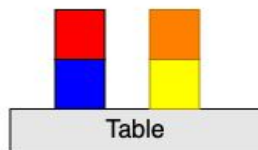
salesforce

REX: UCB-CoT



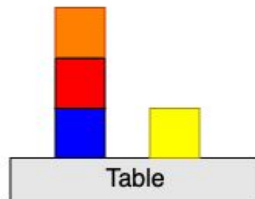
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



Pass-1



REX: UCB-CoT



AI Agent

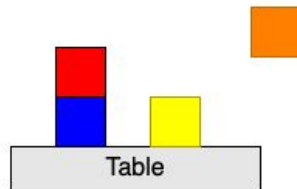
Correct Answer?



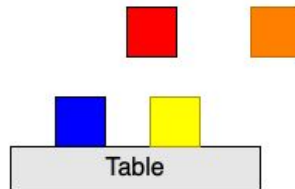
Feedback

Pass-1 Agent's Output:

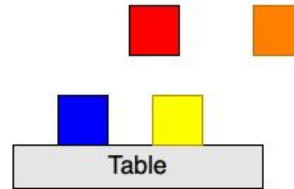
- Step 1: Unstack Orange block from on top of Yellow block
- Step 2: Pick Red block



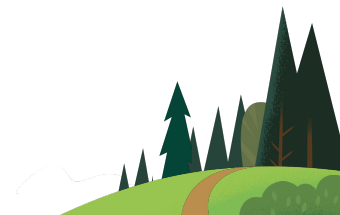
Step 1



Step 2



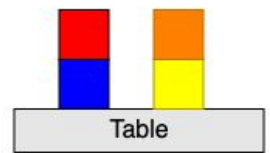
Final Config



REX: UCB-CoT

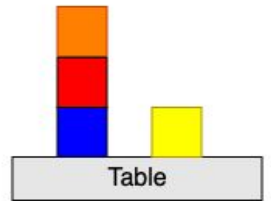
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



Pass-1



AI Agent



UCB

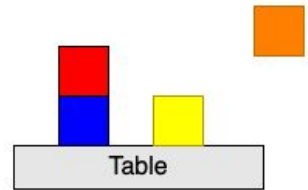
← Feedback →

Correct Answer?

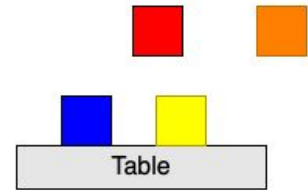


Pass-1 Agent's Output:

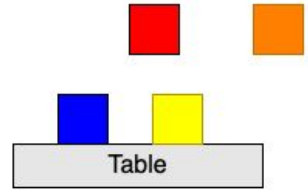
- Step 1:** Unstack Orange block from on top of Yellow block
- Step 2:** Pick Red block



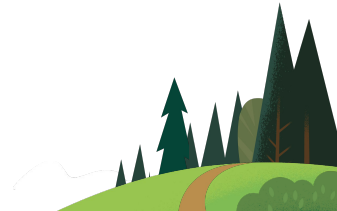
Step 1



Step 2

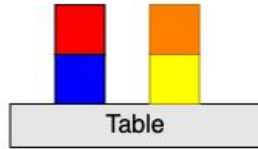


Final Config



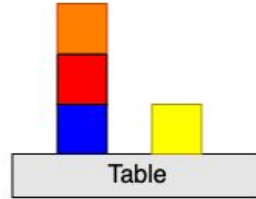
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



AI Agent

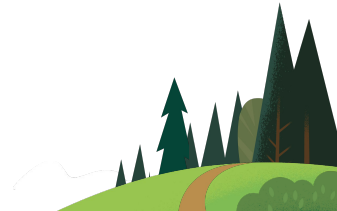


UCB

Pass-2

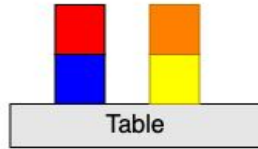
salesforce

REX: UCB-CoT



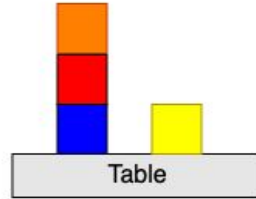
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



AI Agent



UCB



Pass-2 Agent's Output:

Step 1:

Pick Red block

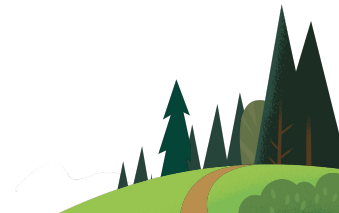
Step 2:

Stack Red block on top of Blue block

Pass-2

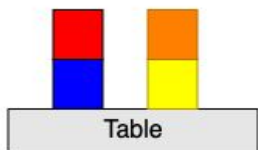
salesforce

REX: UCB-CoT



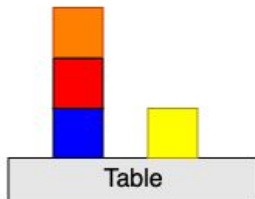
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



UCB



Pass-2 Agent's Output:

Step 1:

Pick Red block

Step 2:

Stack Red block on top of Blue block

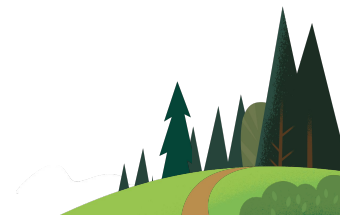
Step 1

Step 2

Pass-2

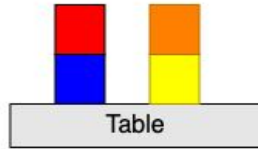


REX: UCB-CoT



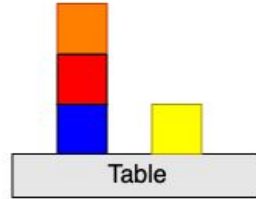
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



UCB



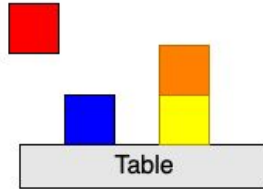
Pass-2 Agent's Output:

Step 1:

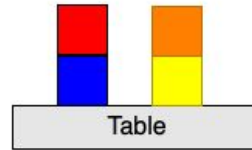
Pick Red block

Step 2:

Stack Red block on top of Blue block



Step 1

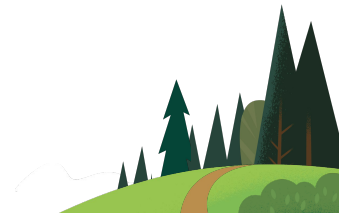


Step 2

Pass-2

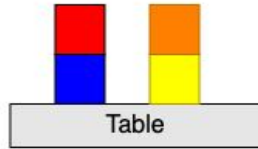
salesforce

REX: UCB-CoT



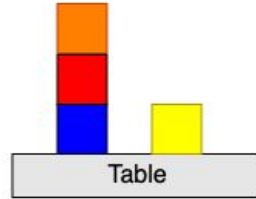
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



UCB



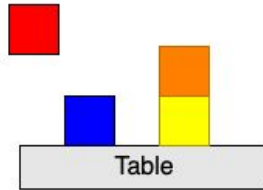
Pass-2 Agent's Output:

Step 1:

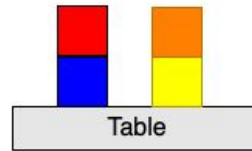
Pick Red block

Step 2:

Stack Red block on top of Blue block



Step 1



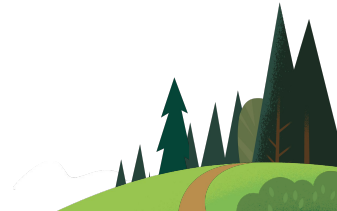
Step 2

Final Config

Pass-2

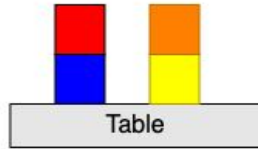
salesforce

REX: UCB-CoT



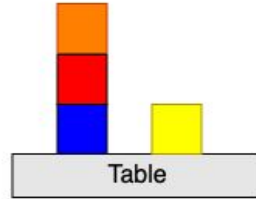
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



UCB



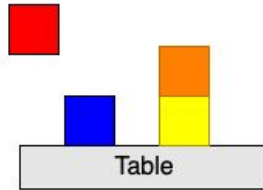
Pass-2 Agent's Output:

Step 1:

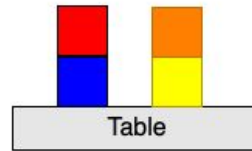
Pick Red block

Step 2:

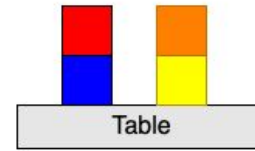
Stack Red block on top of Blue block



Step 1



Step 2

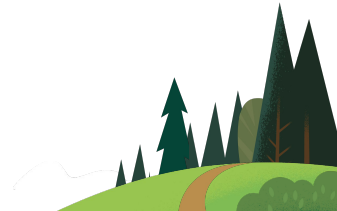


Final Config

Pass-2

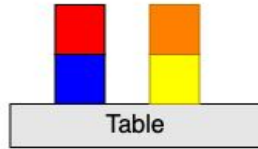
salesforce

REX: UCB-CoT



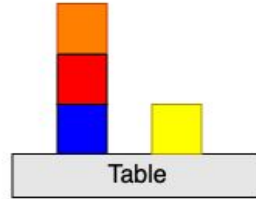
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



Pass-2



REX: UCB-CoT

Correct Answer?



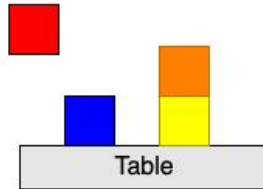
Pass-2 Agent's Output:

Step 1:

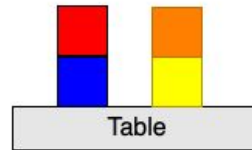
Pick Red block

Step 2:

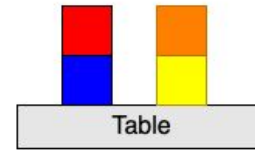
Stack Red block on top of Blue block



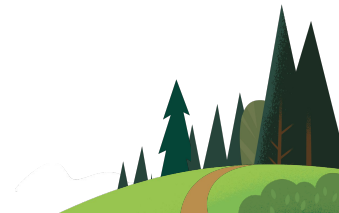
Step 1



Step 2

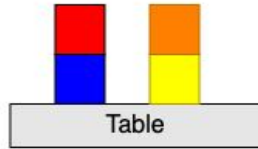


Final Config



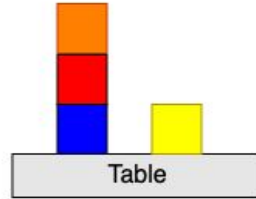
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



UCB



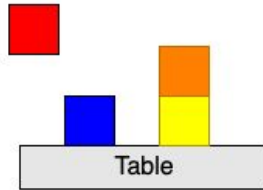
Pass-2 Agent's Output:

Step 1:

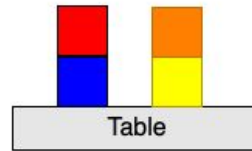
Pick Red block

Step 2:

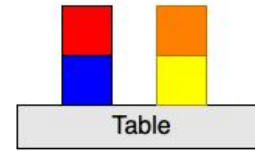
Stack Red block on top of Blue block



Step 1



Step 2



Final Config

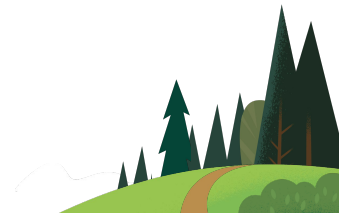
Correct Answer?



Pass-2

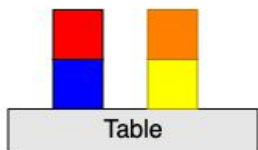
salesforce

REX: UCB-CoT



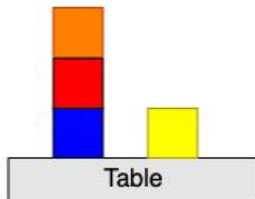
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



Pass-2



REX: UCB-CoT



AI Agent



UCB

Feedback

Correct Answer?



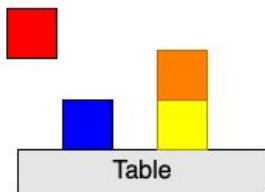
Pass-2 Agent's Output:

Step 1:

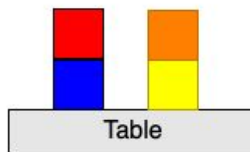
Pick Red block

Step 2:

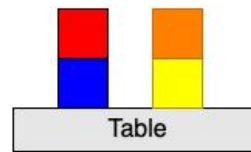
Stack Red block on top of Blue block



Step 1



Step 2

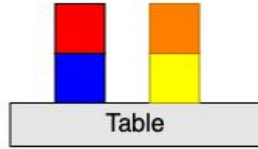


Final Config



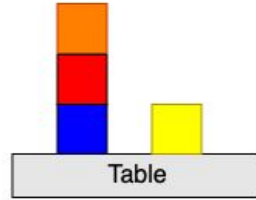
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



AI Agent

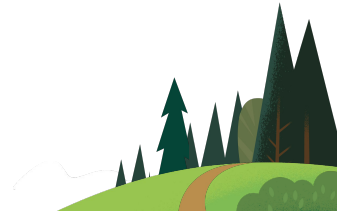


UCB

Pass-7

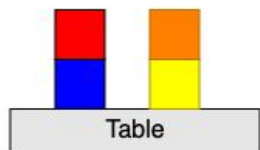
salesforce

REX: UCB-CoT



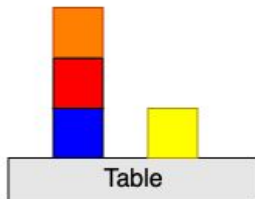
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



AI Agent



UCB



Pass-7 Agent's Output:

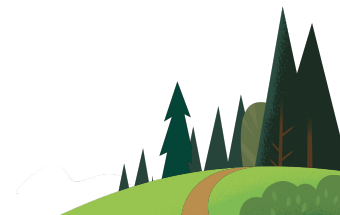
Step 1: Unstack Orange block from on top of Yellow block

Step 2: Stack Orange block on top of Red block

Pass-7

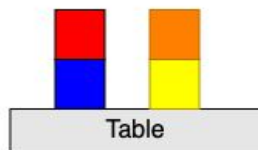


REX: UCB-CoT



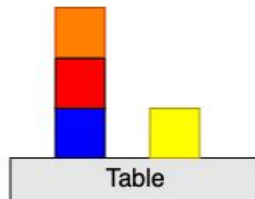
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free



AI Agent



UCB



Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Stack Orange block on top of Red block

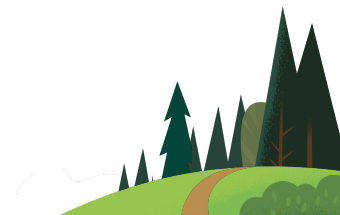
Step 1

Step 2

Pass-7



REX: UCB-CoT



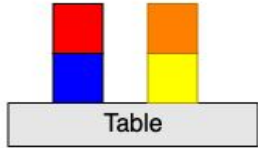
Pass-7



REX: UCB-CoT

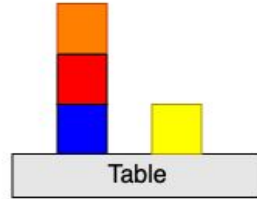
Initial Block Configuration

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Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



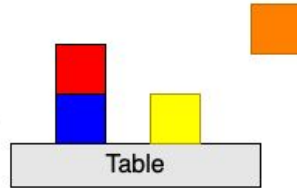
UCB



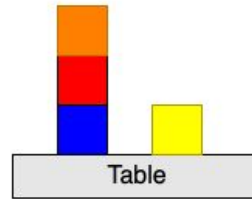
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

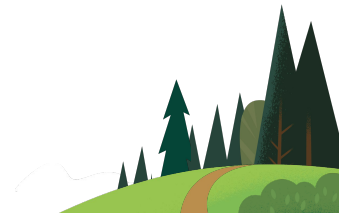
Step 2: Stack Orange block on top of Red block



Step 1

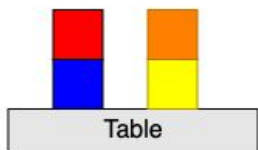


Step 2



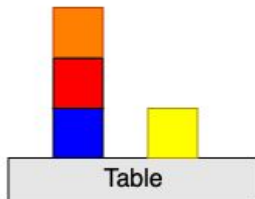
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



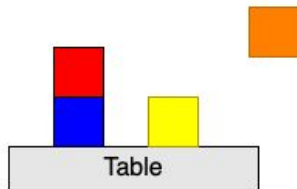
UCB



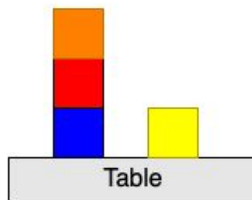
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Stack Orange block on top of Red block



Step 1



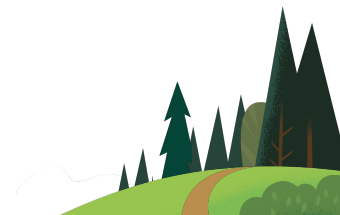
Step 2

Final Config

Pass-7

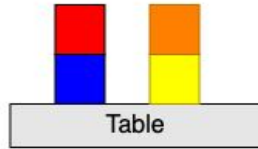


REX: UCB-CoT



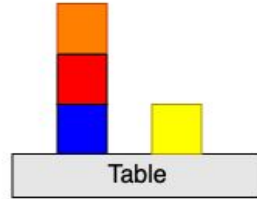
Initial Block Configuration

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Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



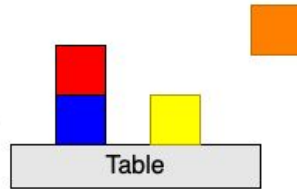
UCB



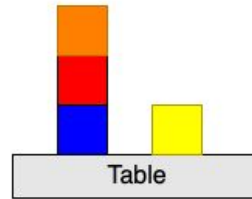
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

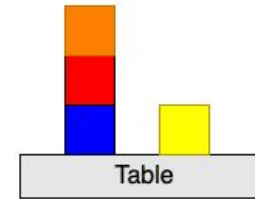
Step 2: Stack Orange block on top of Red block



Step 1



Step 2

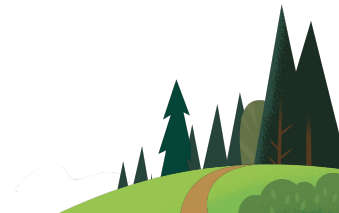


Final Config

Pass-7

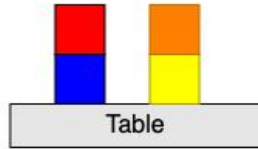
salesforce

REX: UCB-CoT



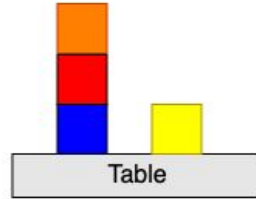
Initial Block Configuration

Blue block is on the table. Red block is on top of Blue block and is free. Yellow block is on table. Orange block is on top of Yellow block and is free.



Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



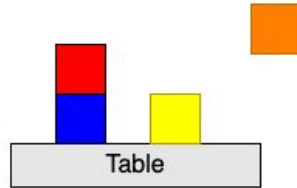
UCB



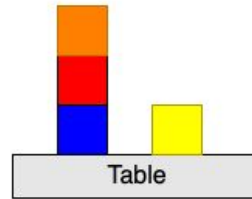
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Stack Orange block on top of Red block

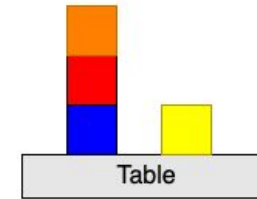


Step 1



Step 2

Correct Answer?

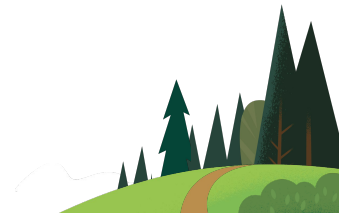


Final Config

Pass-7

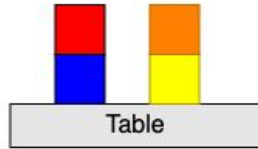
salesforce

REX: UCB-CoT



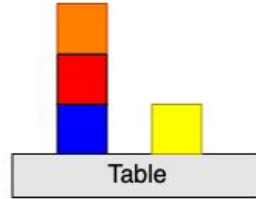
Initial Block Configuration

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Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



AI Agent



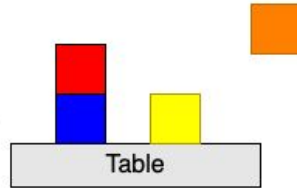
UCB



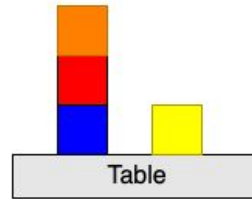
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

Step 2: Stack Orange block on top of Red block

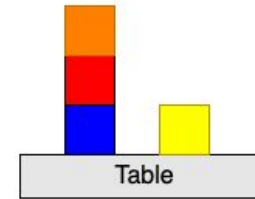


Step 1



Step 2

Correct Answer?

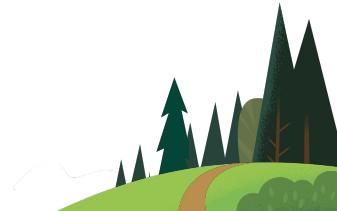


Final Config

Pass-7

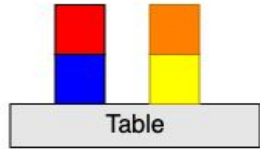
salesforce

REX: UCB-CoT



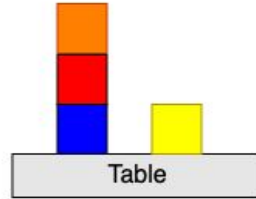
Initial Block Configuration

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Final Block Configuration

Blue block is on the table. Red block is on top of Blue block. Yellow block is on table and is free. Orange block is on top of Red block and is free.



Pass-7



REX: UCB-CoT



AI Agent



UCB

Feedback

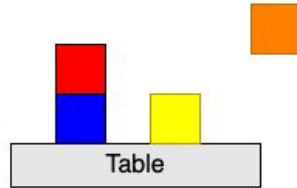
Correct Answer?



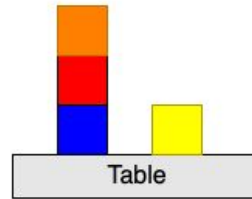
Pass-7 Agent's Output:

Step 1: Unstack Orange block from on top of Yellow block

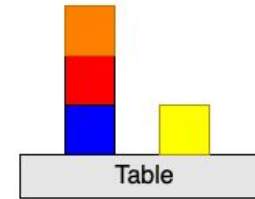
Step 2: Stack Orange block on top of Red block



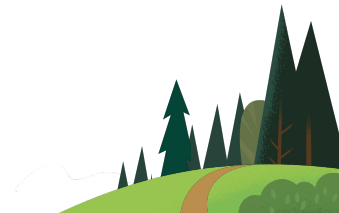
Step 1



Step 2



Final Config



REX Results



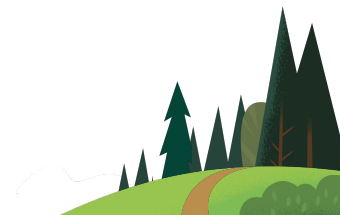
REX {

Model	2-step (n=30)	4-step (n=56)	6-step (n=114)
CoT	40%	17.85%	8.77%
Reflexion (+ CoT)	41.67%	<u>41.96%</u>	29.82%
$\mathcal{R}$ -CoT	53.33%	37.5%	14.91%
UCB-CoT	80%	39.28%	<u>25.43%</u>
UCL-CoT	<u>60%</u>	44.64%	20.17%

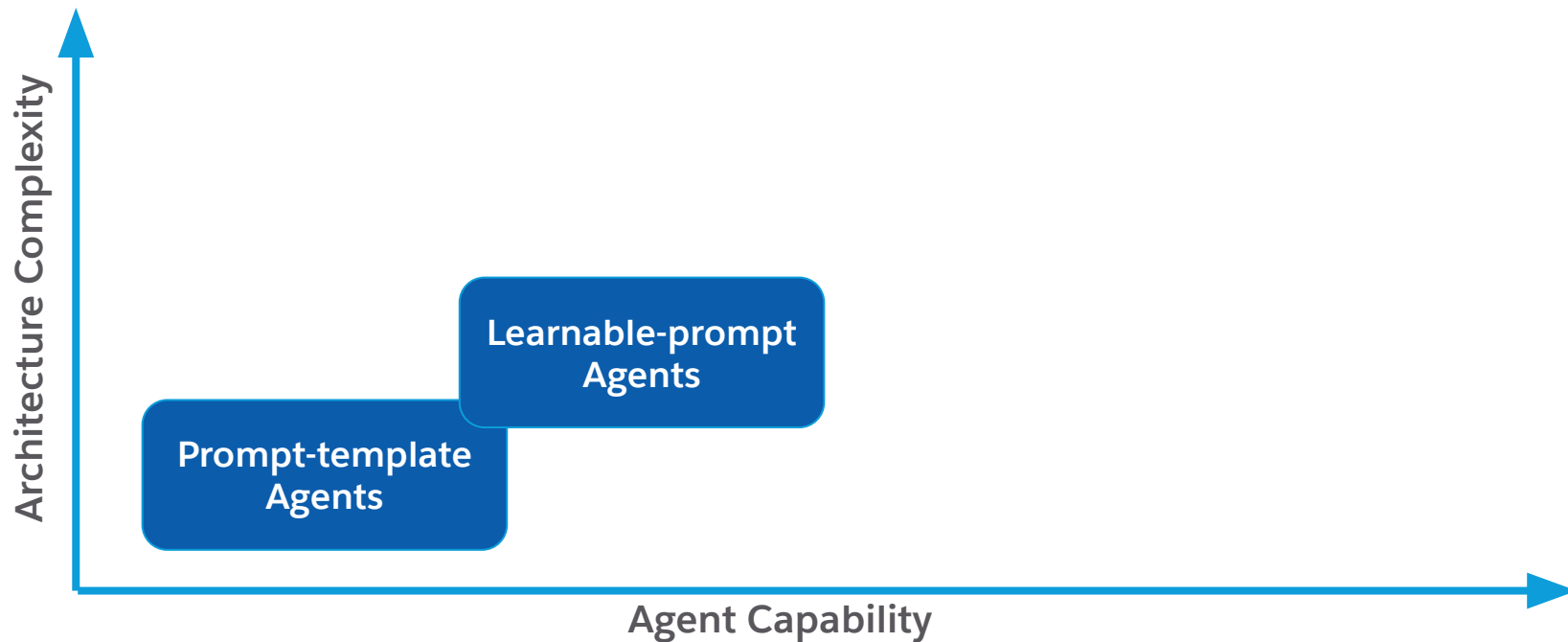
Table 2: Blocksworld; T=0

Model	GSM8K-test (n=1319)
CoT (1-pass)	76.95%
CoT	80.81%
Reflexion (+ CoT)	<u>88.85%</u>
$\mathcal{R}$ -CoT	81.34%
UCB-CoT	82.03%
UCL-CoT	90.44%

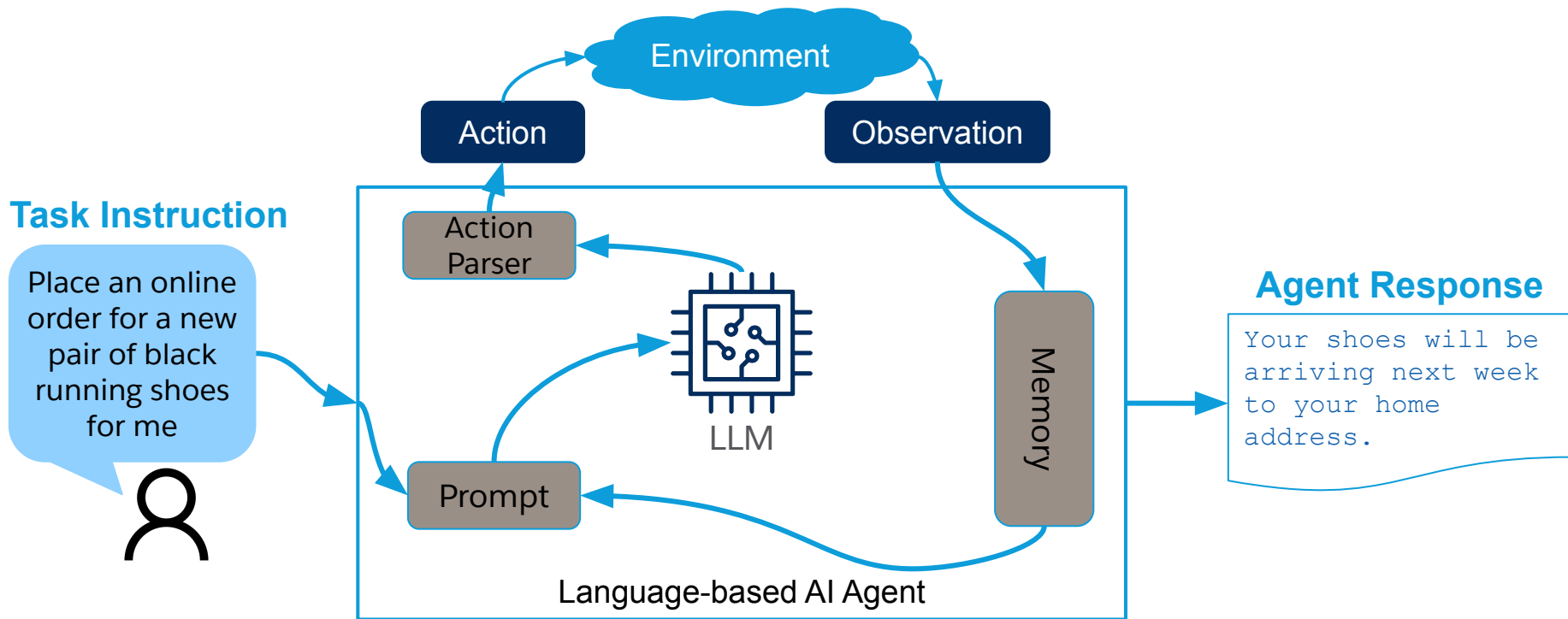
Table 4: GSM8K; T=0



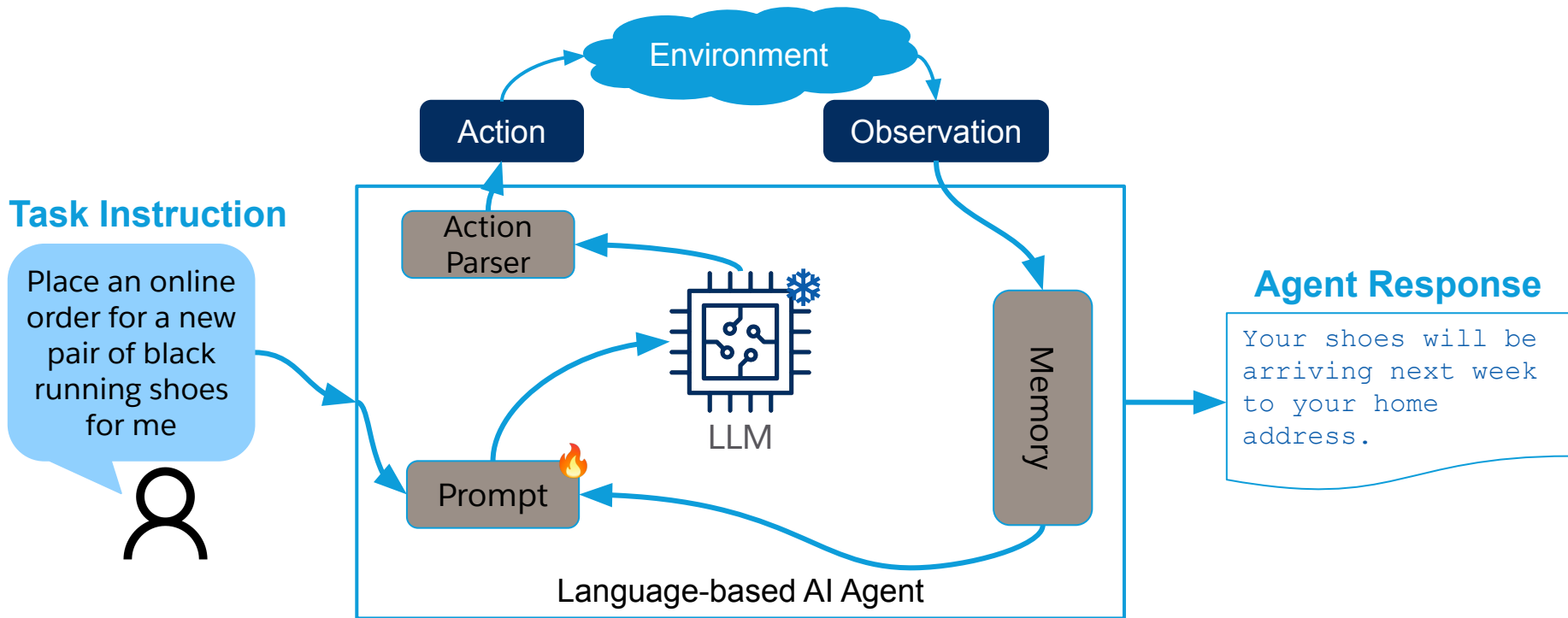
Language-based AI Agents



High-level Language-based Agent framework

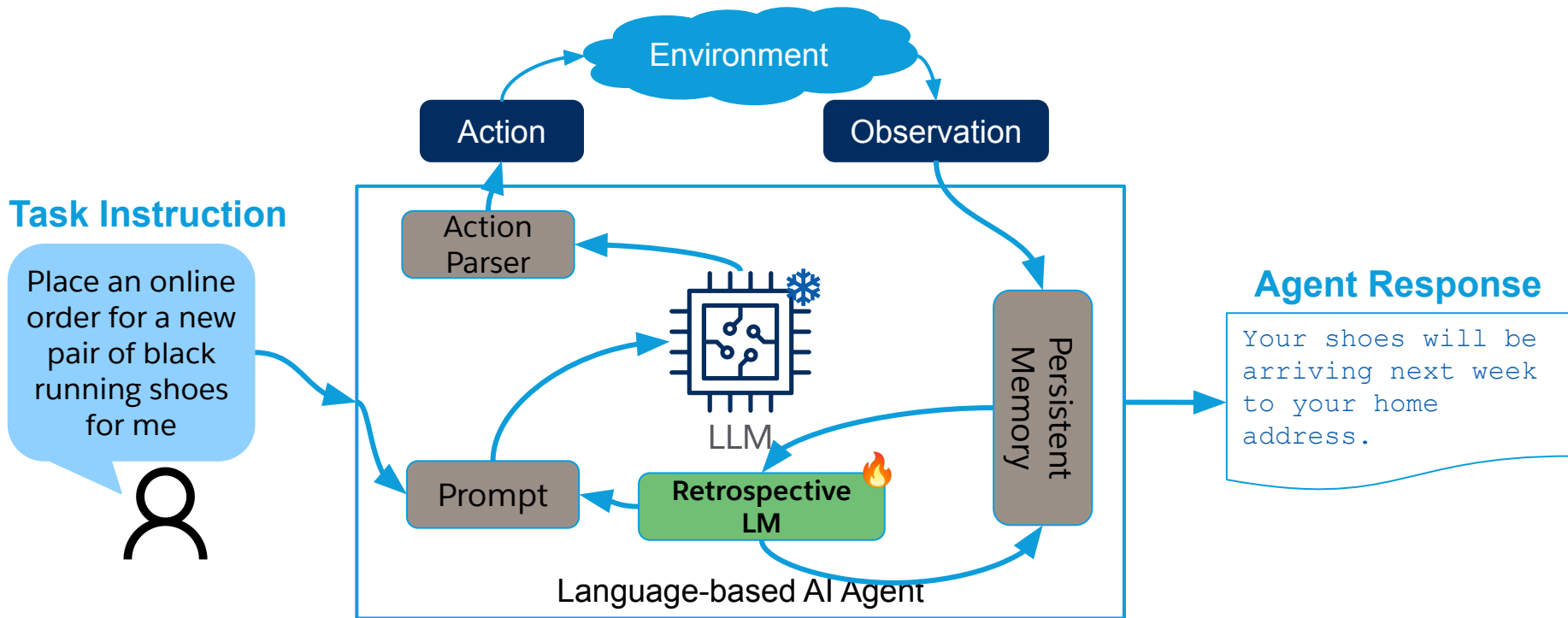


Learnable-prompt Agents



Retroformer

<https://github.com/weirayao/Retroformer>



Retroformer: Store useful reflections

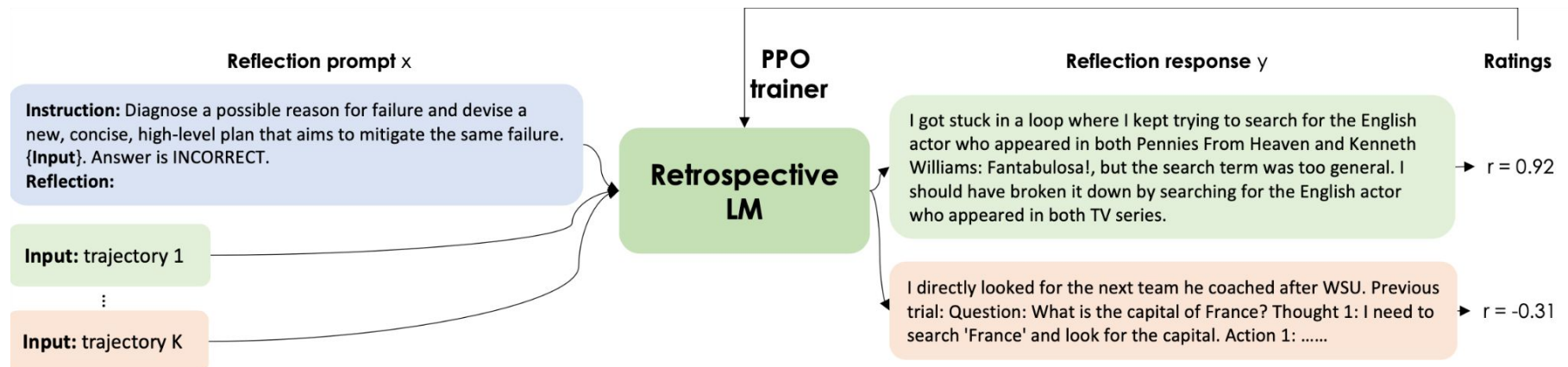


Question	Prior failed trial	Reflection (Green: Retroformer;
Lollipop Chainsaw featured Juliet Starling, who was voiced by a Canadian-American actress who has done voice roles for what Teen Titans spinoff series ?	Thought 1: I need to search Juliet Starling and find the Canadian-American actress who voiced her, then find the Teen Titans spinoff series she has done voice roles for. Action 1: Search[Juliet Starling] Observation 1: Could not find [Juliet Starling]. ...	I failed to find the Teen Titans spinoff series that Tara Strong has done voice roles for. Instead, I found Lollipop Chainsaw, a video game that Juliet Starling appeared in. I should have searched Tara Strong and Teen Titans or Teen Titans Go! specifically in my previous action to find the correct answer.
	Thought 2: Juliet Starling is a character from Lollipop Chainsaw, so I can search Lollipop Chainsaw. Action 2: Search[Lollipop Chainsaw] Observation 2: Lollipop Chainsaw is a ...	
	Thought 3: Juliet Starling is voiced by Tara Strong, so I need to search Tara Strong and find the Teen Titans spinoff series she has done voice roles for. Action 3: Search[Tara Strong] Observation 3: Tara Lyn Strong...	
	Thought 4: Tara Strong has voiced characters in Teen Titans and Teen Titans Go!, so the answer is Teen Titans and Teen Titans Go!. Action 4: Finish[Teen Titans and Teen Titans Go!]	

Retroformer: Policy Gradient Optimization



Use Proximal Policy Optimization (PPO) to fine-tune the retrospective LM



Retroformer: HotPotQA Result

Question Answering Task

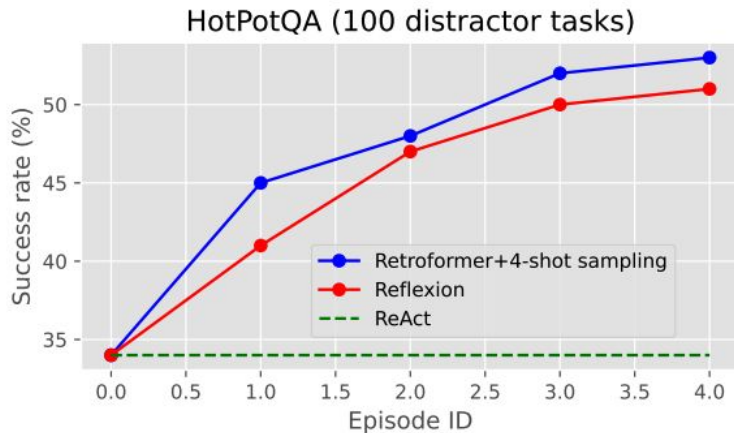
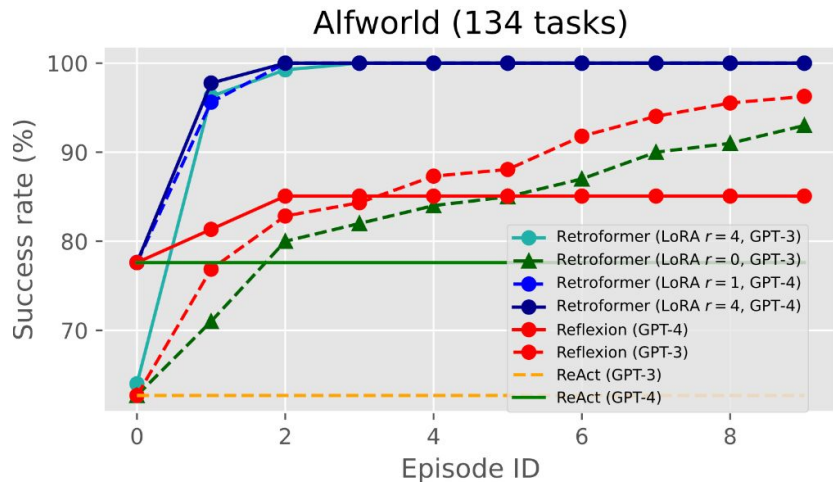


Figure 4: We compare **Retroformer** with baselines and observe consistent and faster performance improvement in terms of success rate.

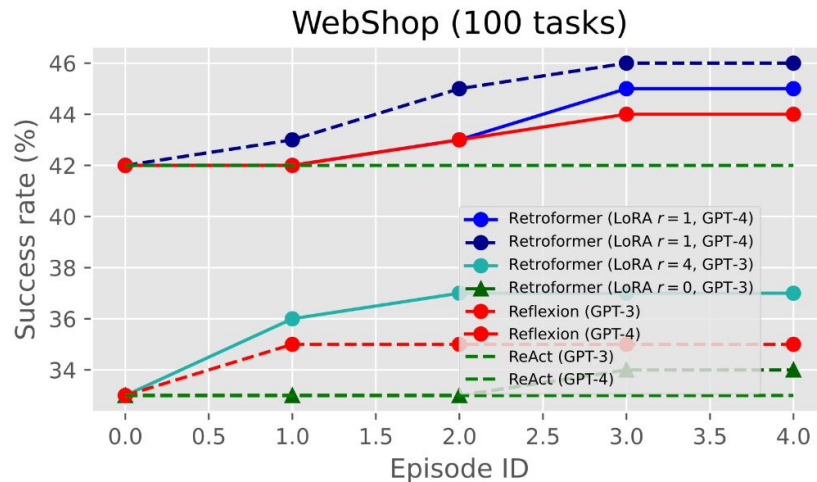
Task Description

- Solve a question answering task by searching in Wikipedia pages.
- At each step, the agent is asked to choose from three action APIs:
 - `Search[entity]`
 - `Lookup[keyword]`
 - `Finish[answer]`
- Evaluation metric (success rate) measures if the answer and ground truth matches exactly or not, over all tasks.

Retroformer: Alfworld & Webshop Results



(a) AlfWorld environment

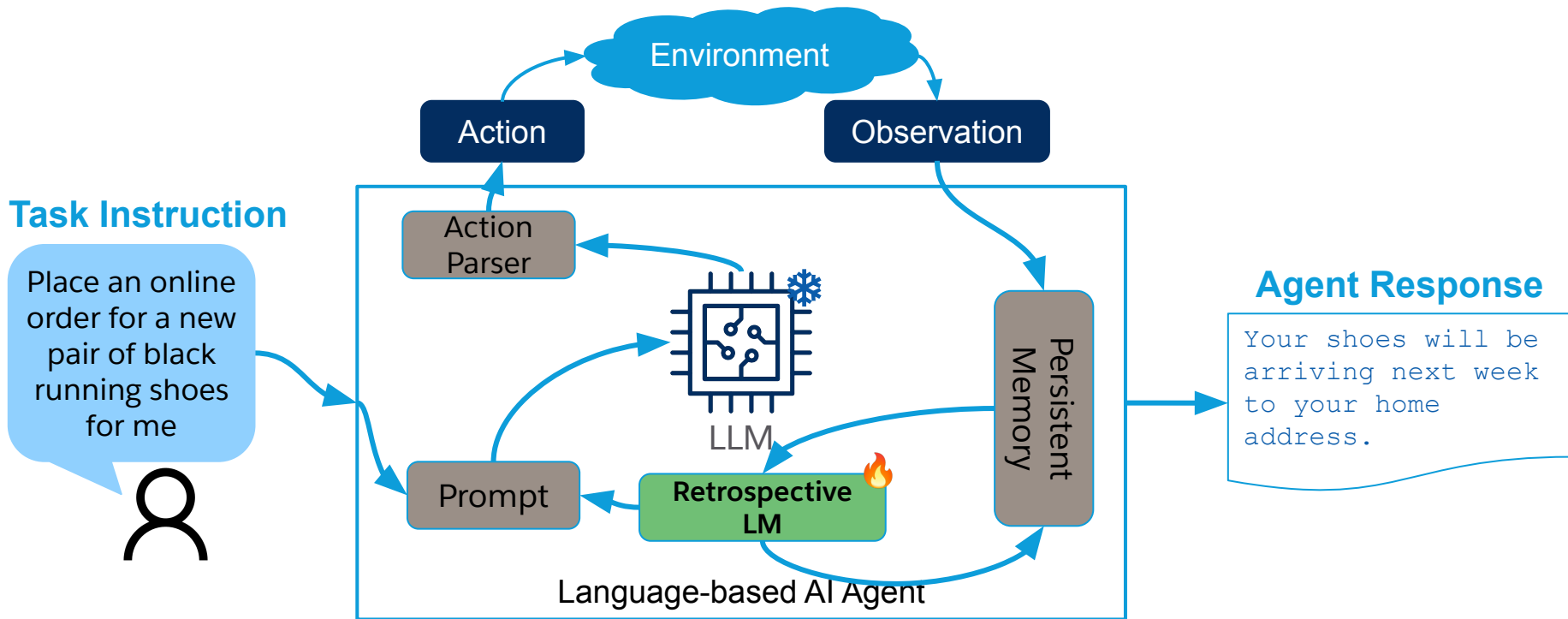


(b) WebShop environment

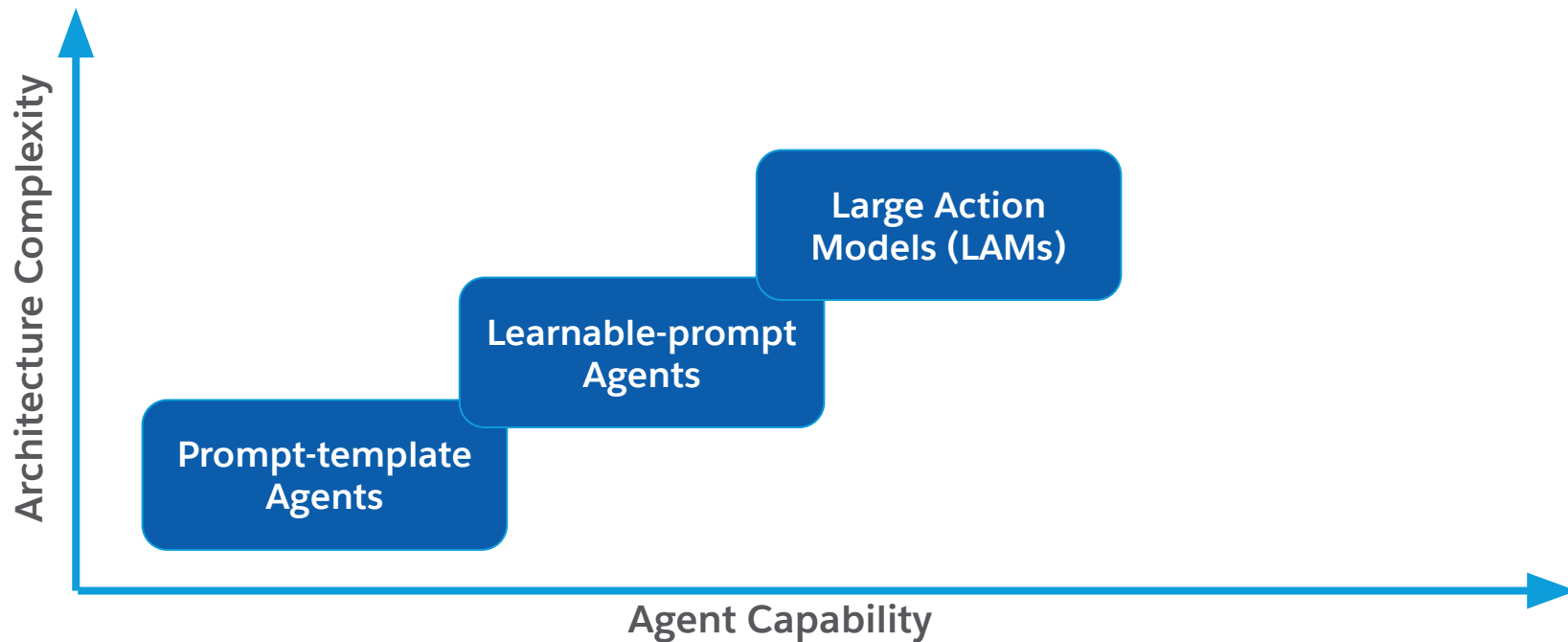
Retroformer takeaways: Refine prompts, decompose



<https://github.com/weirayao/Retroformer>

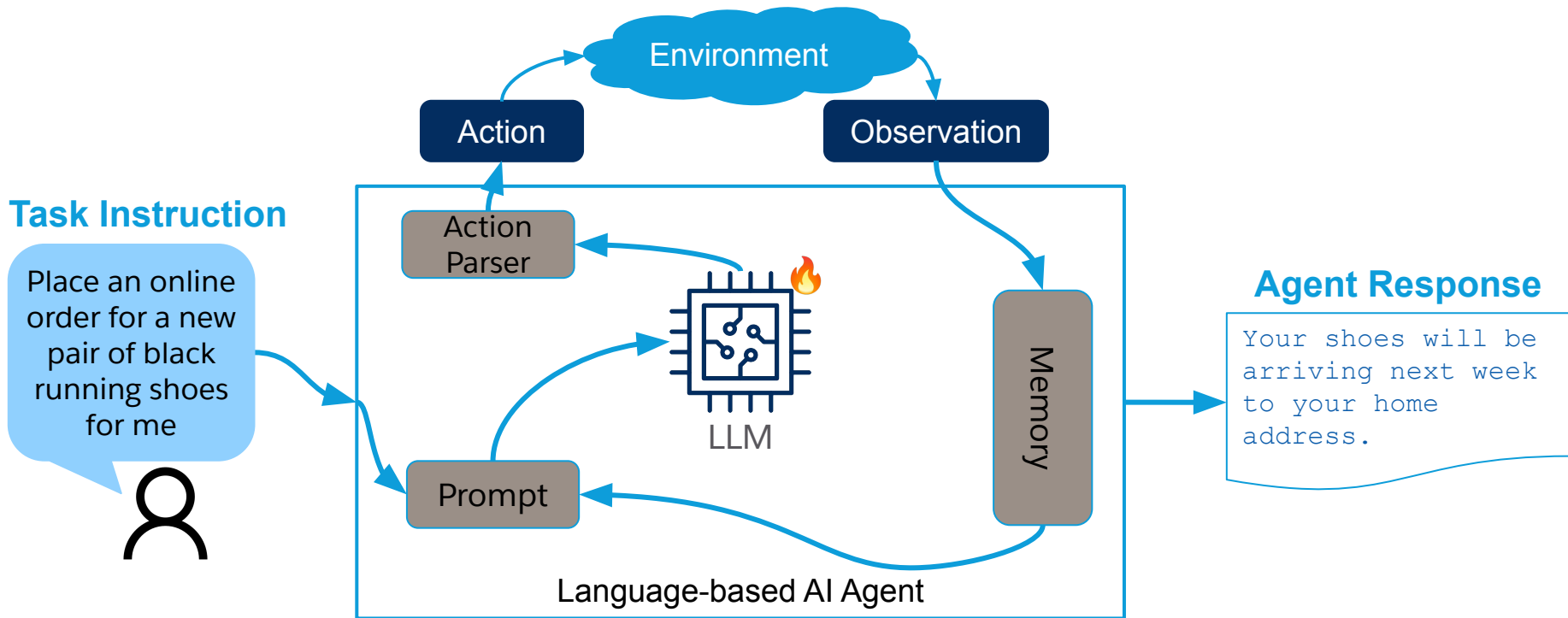


Language-based AI Agents



Large Action Models (LAMs)

<https://github.com/SalesforceAIResearch/xLAM>



[Zhang et al. The AgentOhana: Designing Unified Data and Training Pipeline for Effective Agent Learning. ICLR 2024 Workshop on LLM Agents]

Agent Trajectory Data Example



HotPotQA

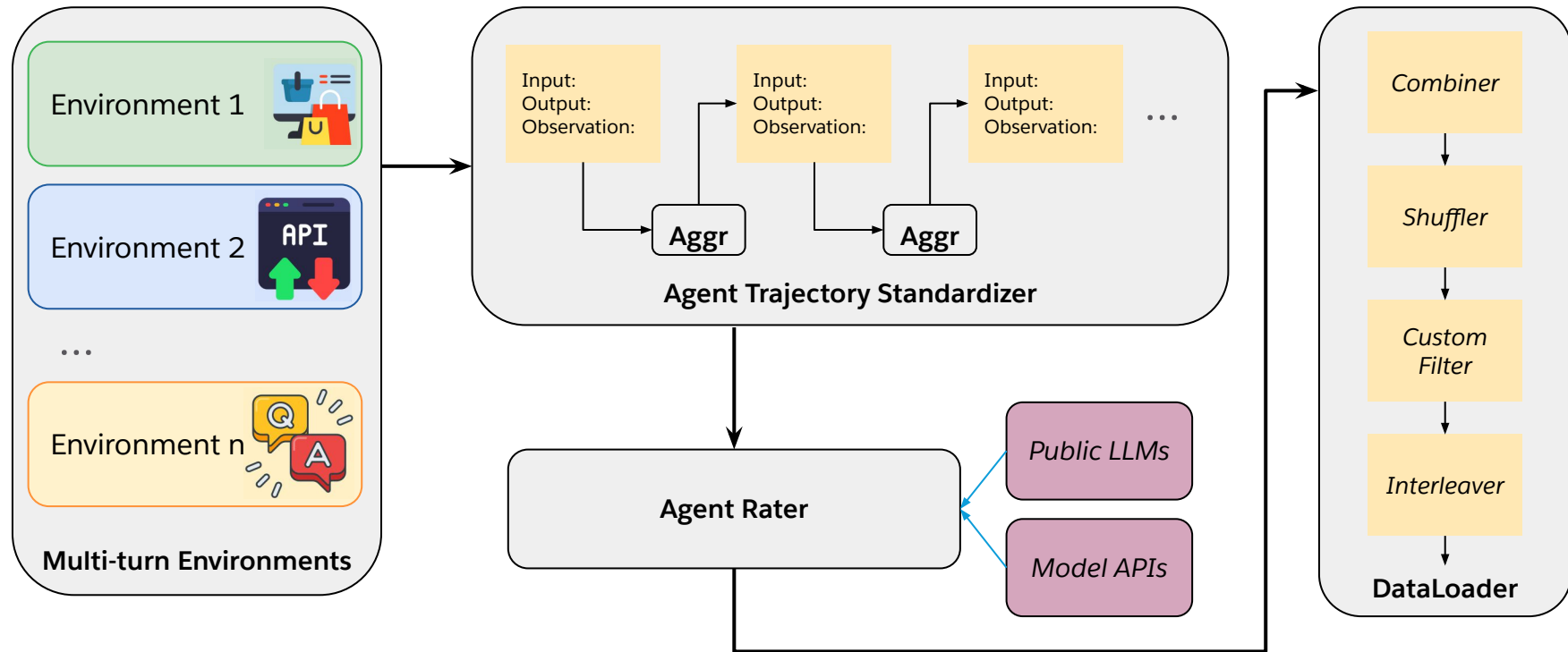
```
{
  "question": "Which magazine was started first Arthur's Magazine or First for Women?",
  "answer": "Arthur's Magazine",
  "correct": true,
  "reward": 1.0,
  "halted": false,
  "error": false,
  "prompt": "
    Question: Which magazine was started first Arthur's Magazine or First for Women? Follow Action[param] format.\n
    Thought: I need to Search[Arthur's Magazine] and then Search[First for Women] to collect their starting information.\n
    Model Action 1: Search[Arthur's Magazine]\n
    Env Observation 1: Arthur's Magazine was published in 1844. \n
    Model Action 2: Search[First for Women]\n
    Env Observation 2: First for Women is a woman's magazine published in 1989. \n
    Model Action 3: Finish[Arthur's Magazine]\n
    Env Observation 3: Answer is CORRECT"
}
```

WebShop

```
[[
  "Begin!\n\nUSER: Can you send a GET request to link 1 and then retrieve the cookie by sending another GET request to link 2\n Follow: Thought: xxx.\n Action: name \n Action Input:{param} format.",
  "ASSISTANT Thought: I need to send a request to set the cookie. \nASSISTANT Action: sendHttpRequest\nASSISTANT Action Input: {link1}\nASSISTANT Observation: ",
  "Status Code: 200.... Response 1\n",
  "ASSISTANT Thought: The request was successful.\nASSISTANT Action: getClientRequestData\nASSISTANT Action Input: {link2}\nASSISTANT Observation: ",
  "Status Code: 200. ... \nASSISTANT Thought: ",
  "The cookie data was retrieved.\nASSISTANT Response: The request was successful."
],
[ false, true, false, true, false, true]
]]
```

AgentOhana: Unified Diverse Training Data

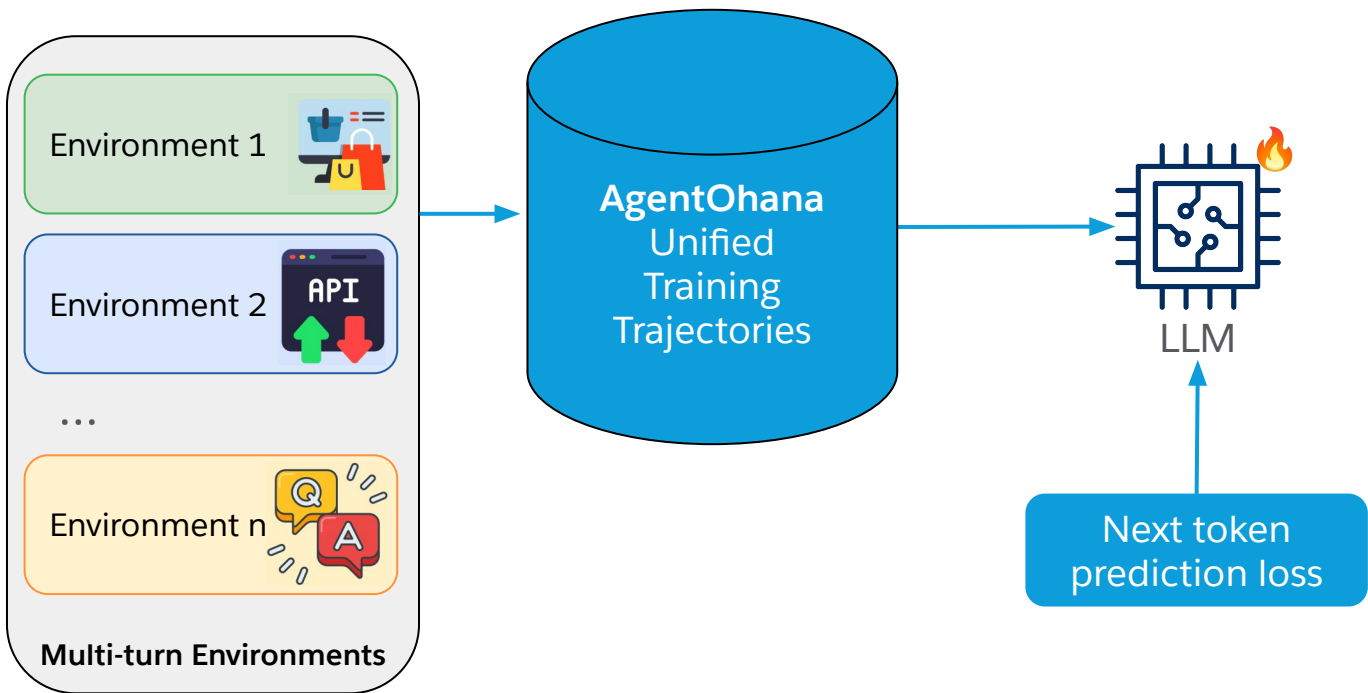
<https://github.com/SalesforceAIResearch/xLAM>



[Zhang et al. The AgentOhana: Designing Unified Data and Training Pipeline for Effective Agent Learning. ICLR 2024 Workshop on LLM Agents]

Training xLAM (Large Action Model)

Supervised Fine-tuning (SFT)



[Zhang et al. The AgentOhana: Designing Unified Data and Training Pipeline for Effective Agent Learning. ICLR 2024 Workshop on LLM Agents]

xLAM: Results



<https://github.com/SalesforceAIResearch/xLAM>

LLM	LAA Architecture					
	ZS	ZST	ReAct	PlanAct	PlanReAct	BOLAA
Llama-2-70b-chat (Touvron et al., 2023)	0.0089	0.0102	0.4273	0.2809	0.3966	0.4986
Vicuna-33b (Zheng et al., 2023b)	0.1527	0.2122	0.1971	0.3766	0.4032	0.5618
Mixtral-8x7B-Instruct-v0.1 (Jiang et al., 2024)	0.4634	0.4592	<u>0.5638</u>	0.4738	0.3339	0.5342
GPT-3.5-Turbo	0.4851	<u>0.5058</u>	0.5047	0.4930	<u>0.5436</u>	<u>0.6354</u>
GPT-3.5-Turbo-Instruct	0.3785	0.4195	0.4377	0.3604	0.4851	0.5811
GPT-4-0613	<u>0.5002</u>	0.4783	0.4616	0.7950	0.4635	0.6129
xLAM-v0.1	0.5201	0.5268	0.6486	<u>0.6573</u>	0.6611	0.6556

Table 2: Average reward on the WebShop environment. **Bold** and Underline results denote the best result and the second best result for each setting, respectively.

[Zhang et al. The AgentOhana: Designing Unified Data and Training Pipeline for Effective Agent Learning. ICLR 2024 Workshop on LLM Agents]

xLAM: Results

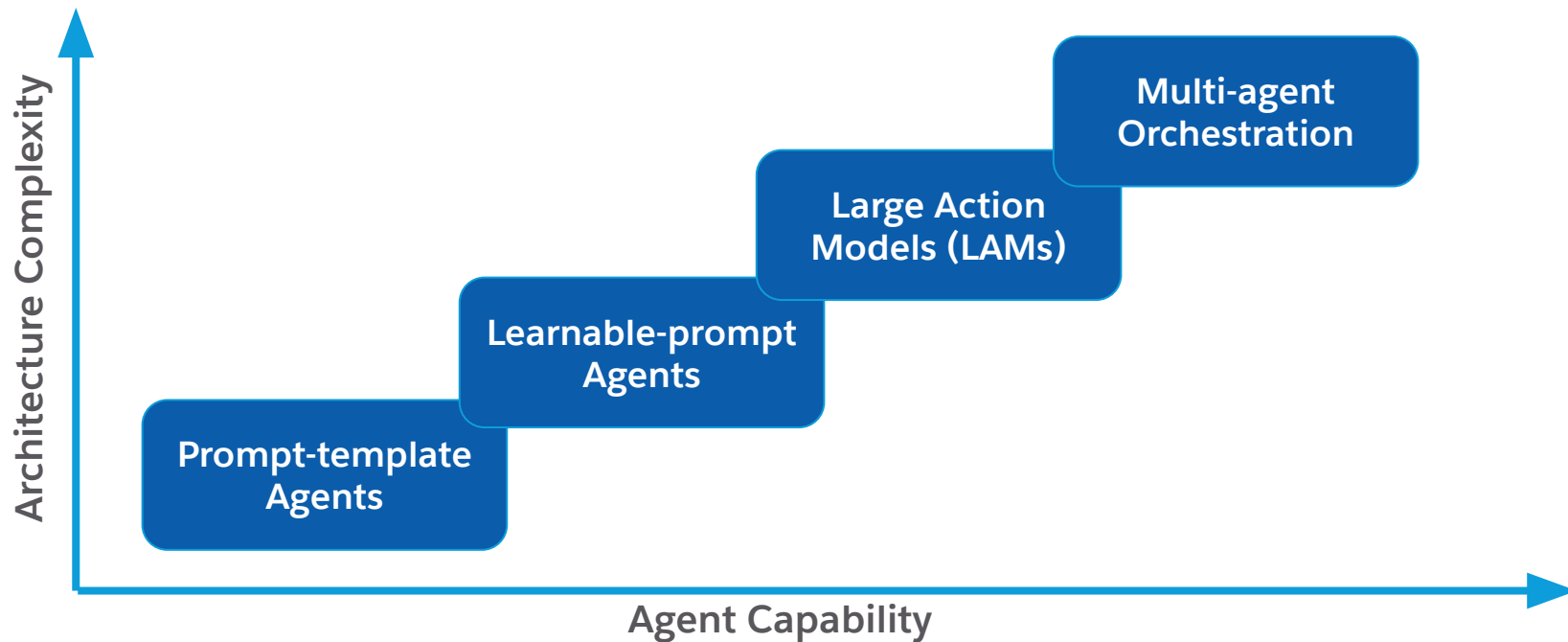


<https://github.com/SalesforceAIResearch/xLAM>

LLM	LAA Architecture				
	ZS	ZST	ReAct	PlanAct	PlanReAct
Mixtral-8x7B-Instruct-v0.1 (Jiang et al., 2024)	0.3912	0.3971	0.3714	0.3195	0.3039
GPT-3.5-Turbo	0.4196	0.3937	0.3868	0.4182	0.3960
GPT-4-0613	0.5801	0.5709	0.6129	0.5778	0.5716
xLAM-v0.1	<u>0.5492</u>	<u>0.4776</u>	<u>0.5020</u>	<u>0.5583</u>	<u>0.5030</u>

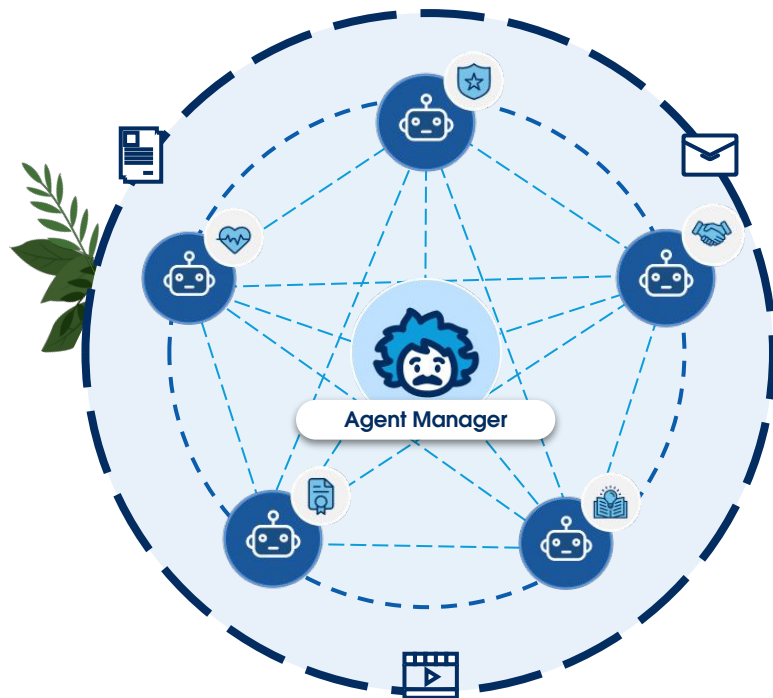
Table 3: Average reward on the HotpotQA environment. **Bold** and Underline results denote the best result and the second best result for each setting, respectively.

Language-based AI Agents



From Single Agents to Multiple Agents

<https://github.com/SalesforceAIResearch/AgentLite>



release v0.1.0 Python 3.9+ License Apache Stars 244

Equip agents with LLMs and Tools

Define agent hierarchy: managers and specialists

Lean implementation, easy-to-follow tutorials

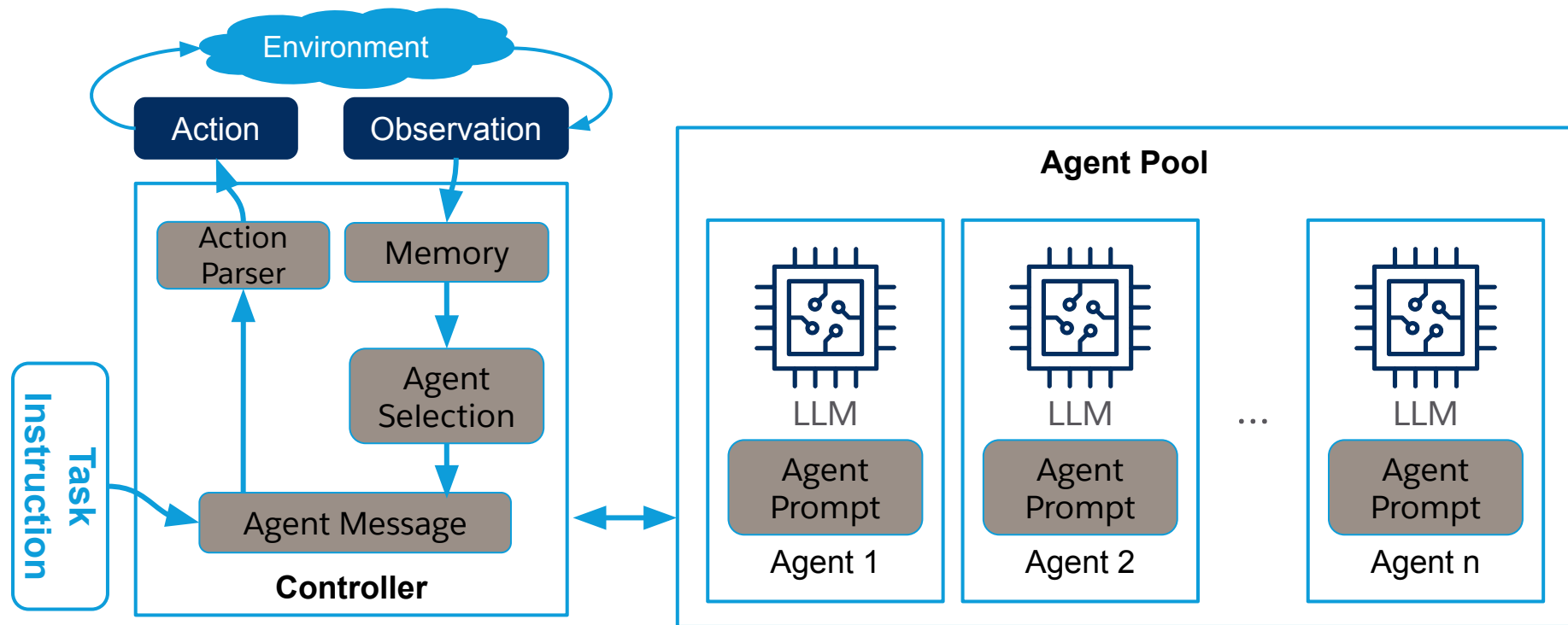
[Liu. AgentLite: A Lightweight Library for Building and Advancing Task-Oriented LLM Agent System. 2024]



BOLAA: Orchestrating multiple agents



<https://github.com/salesforce/BOLAA>



[Zhiwei Liu. "BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents". ICLR 2024 Workshop on LLM Agents]

BOLAA Multi-Agent WebShop Results

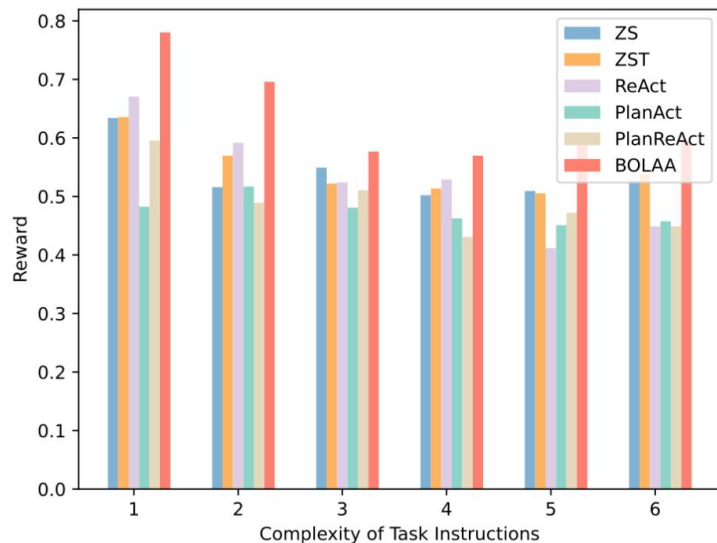


LLM	Length	Zero-Shot	ReAct	BOLAA
fastchat-t5-3b	2k	0.40	0.31	0.52
vicuna-7b	2k	0.00	0.10	0.06
vicuna-13b	2k	0.03	0.15	0.54
vicuna-33b	2k	0.14	0.19	0.56
llama-2-7b	4k	0.00	0.12	0.46
llama-2-13b	4k	0.07	0.26	0.37
llama-2-70b	4k	0.01	0.44	0.50
mpt-7b-instruct	8k	0.00	0.06	0.06
mpt-30b-instruct	8k	0.17	0.31	0.44
xgen-8k-7b-instruct	8k	0.00	0.07	0.37
longchat-7b-16k	16k	0.02	0.07	0.20
longchat-13b-16k	16k	0.00	0.24	0.32
text-davinci-003	4k	0.53	0.55	0.63
gpt-3.5-turbo	4k	0.51	0.54	0.66
gpt-3.5-turbo-16k	16k	0.57	0.49	0.65

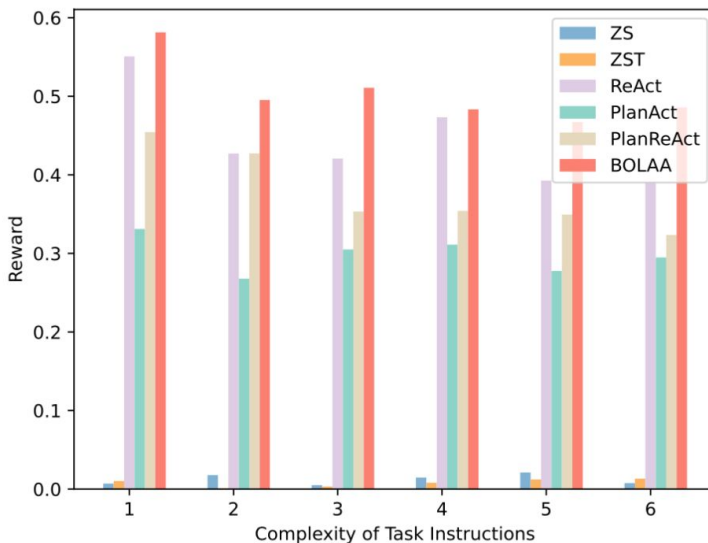
- The BOLAA multi-agent architecture significantly improves the performance of smaller models.
- GPT models also improve

[Zhiwei Liu. “BOLAA: Benchmarking and orchestrating LLM-augmented autonomous agents”. ICLR 2024 Workshop on LLM Agents]

BOLAA: Task Instruction Complexity on WebShop



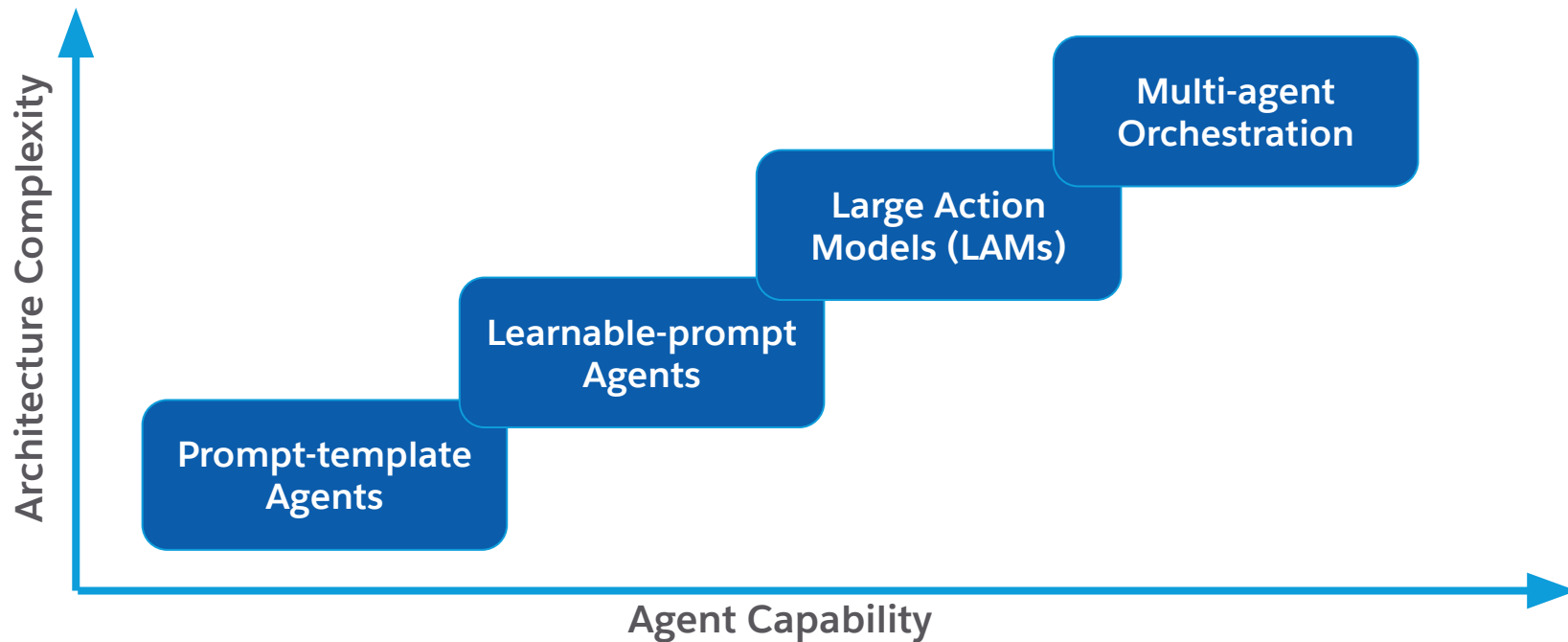
(a) text-davinci-003



(b) Llama-2-70b

- Task Complexity = number of attributes described in query

Language-based AI Agents

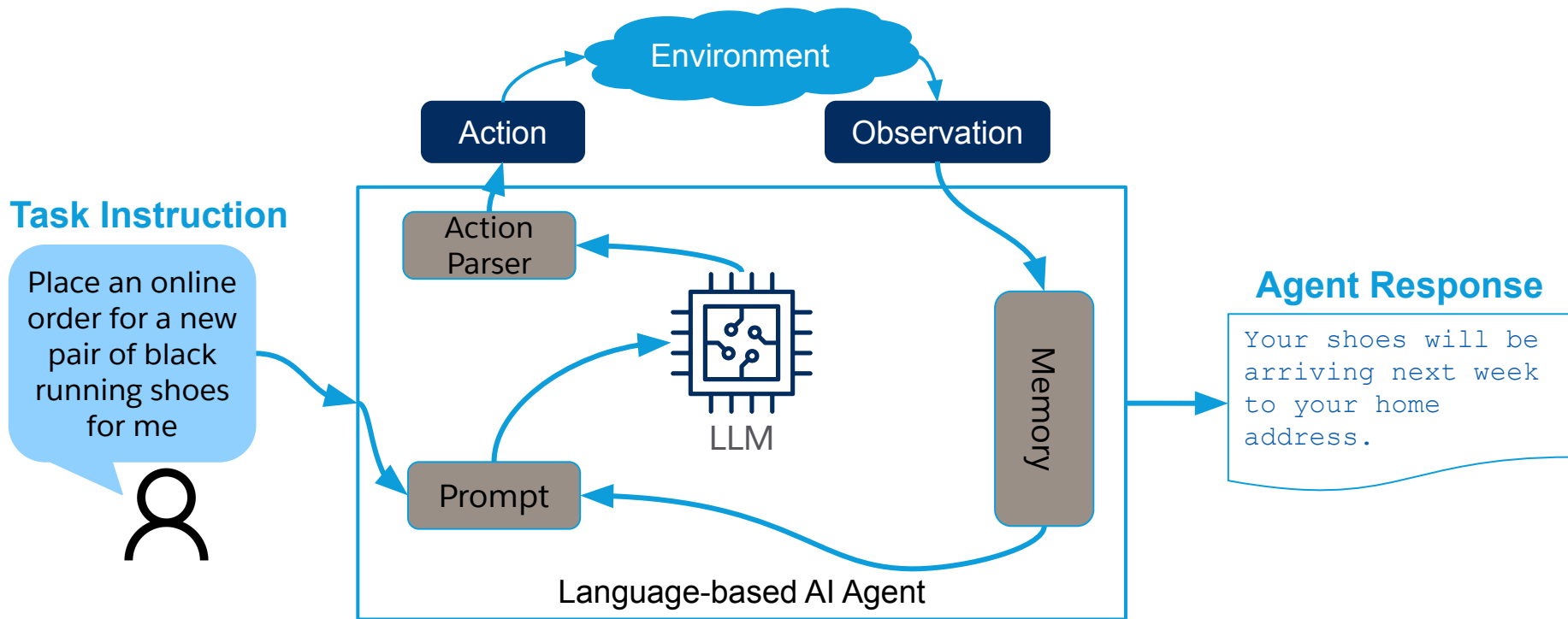


What's next?

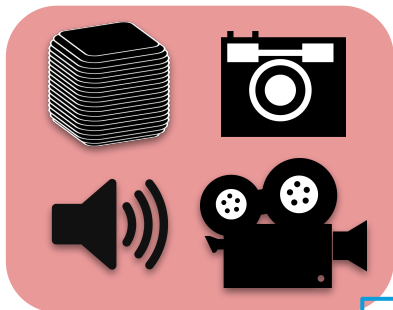
Interesting avenues to further explore



High-level Language-based Agent framework

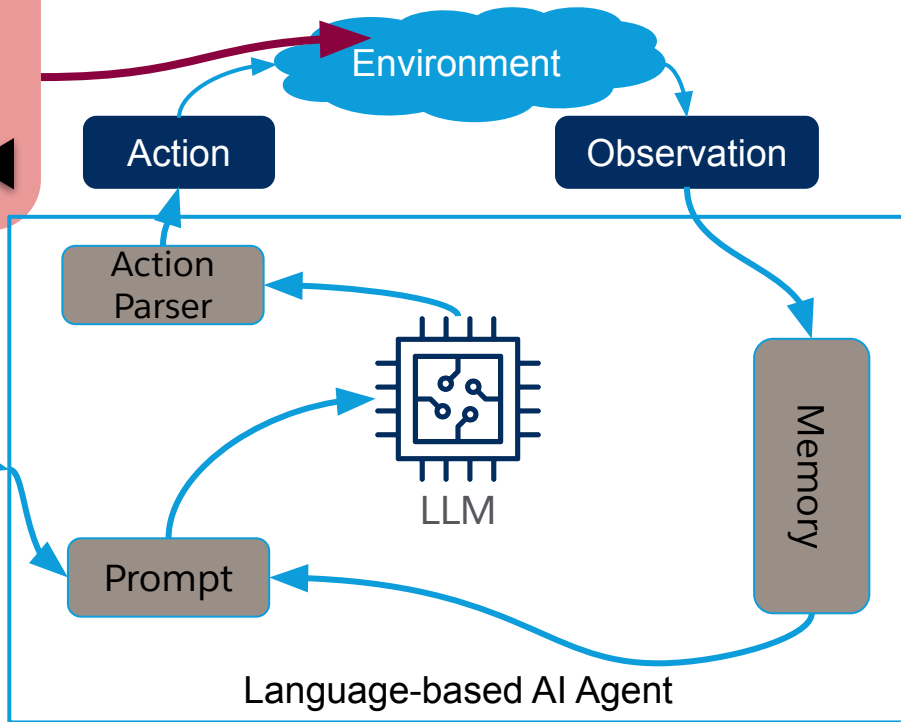


Multimodality



Task Instruction

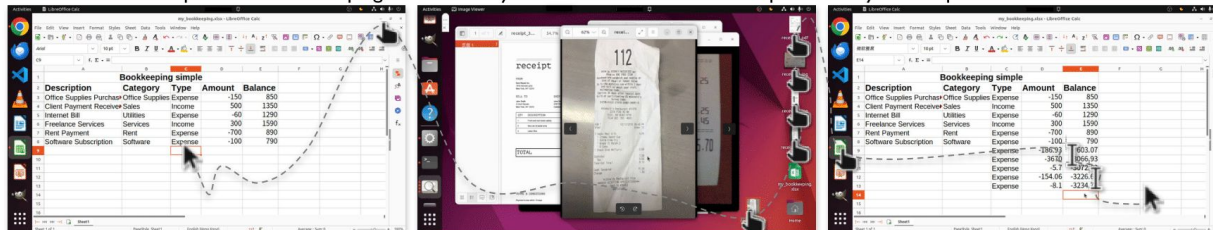
Place an online order for a new pair of black running shoes for me



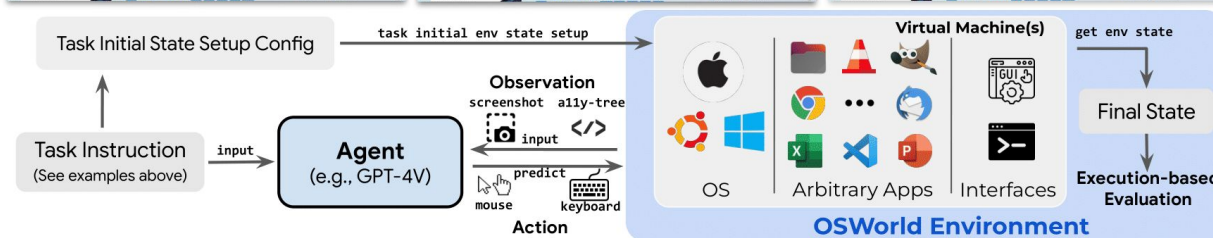
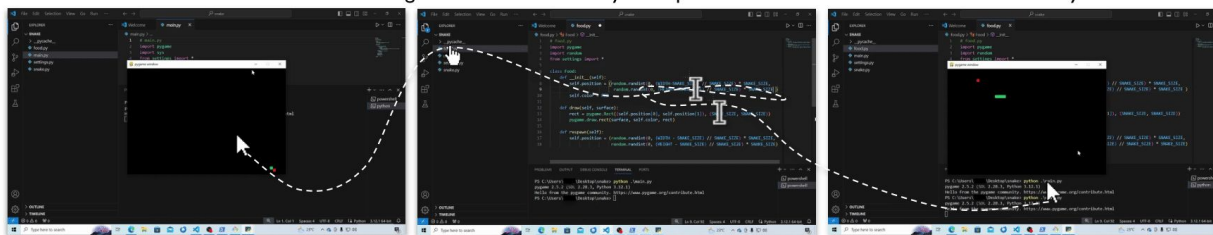
Agent Response

Your shoes will be arriving next week to your home address.

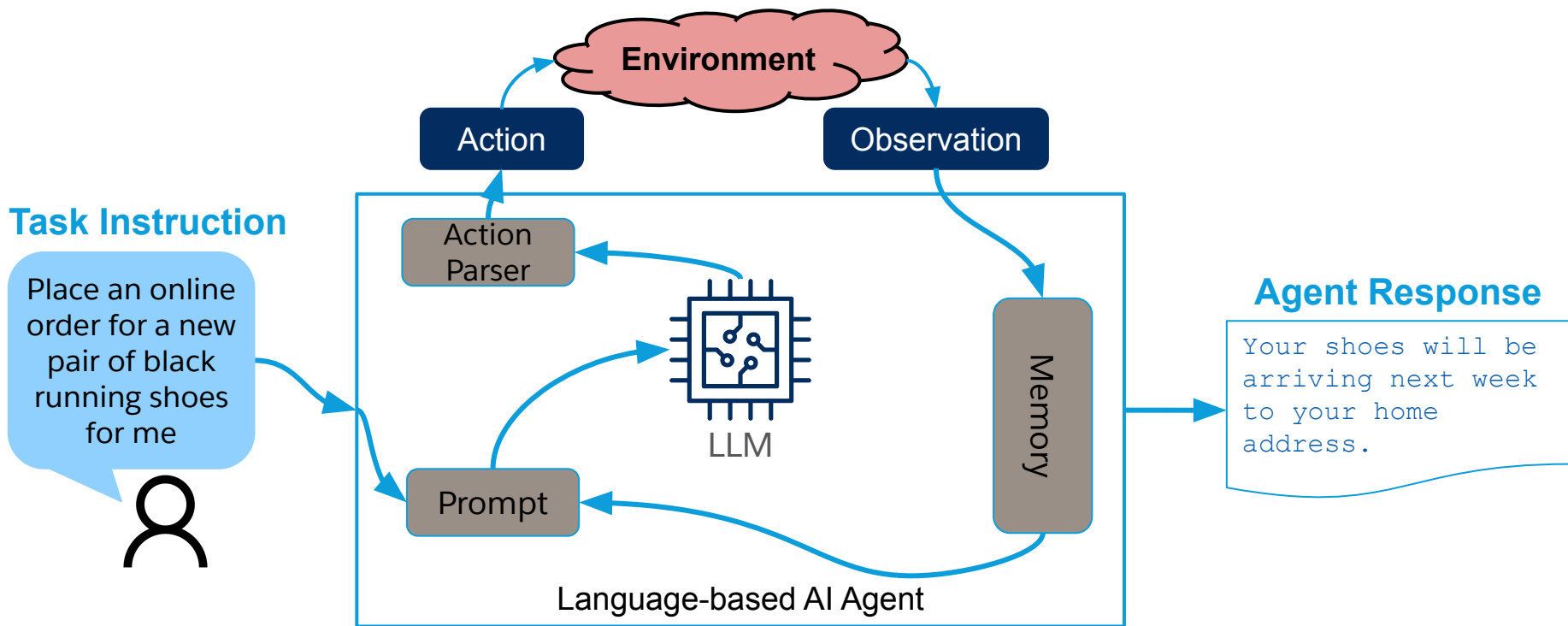
Task instruction 1: Update the bookkeeping sheet with my recent transactions over the past few days in the provided folder.



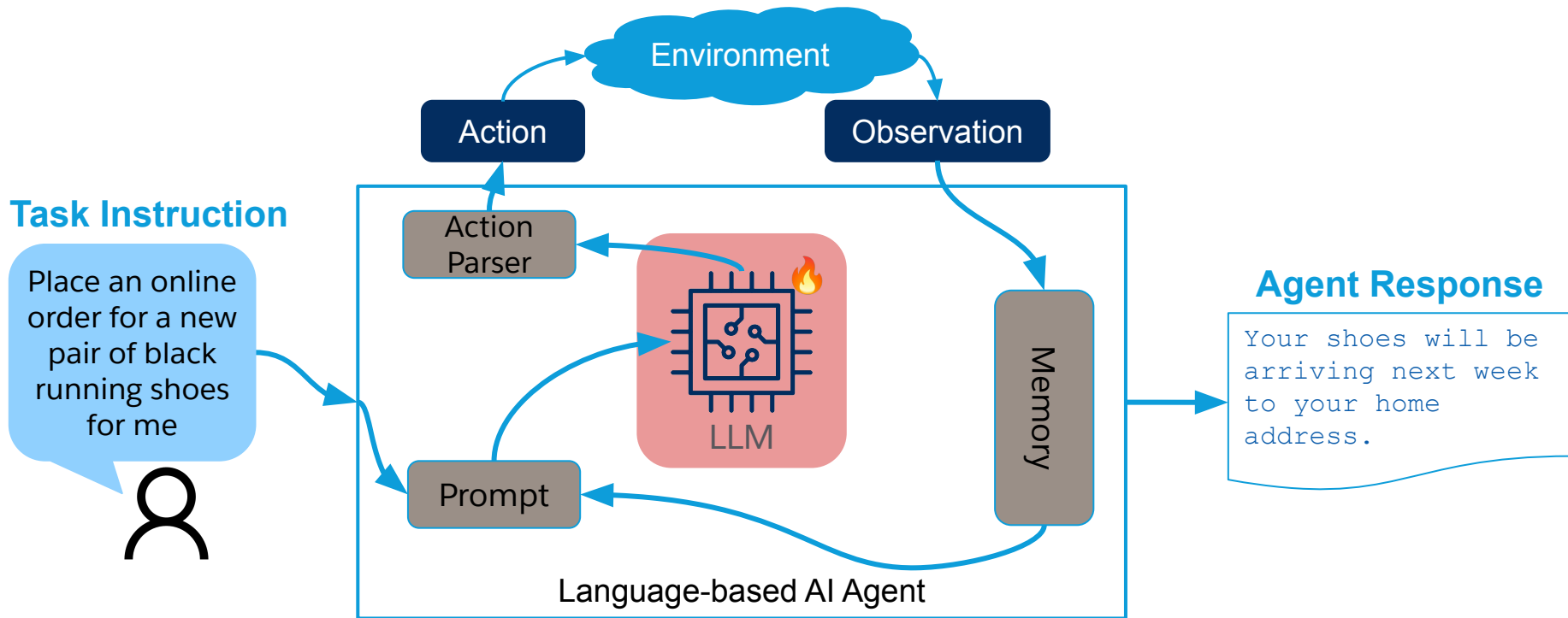
Task instruction 2: ...some details about snake game omitted... Could you help me tweak the code so the snake can actually eat the food?



Benchmarking & Environments



Learning & Generalization



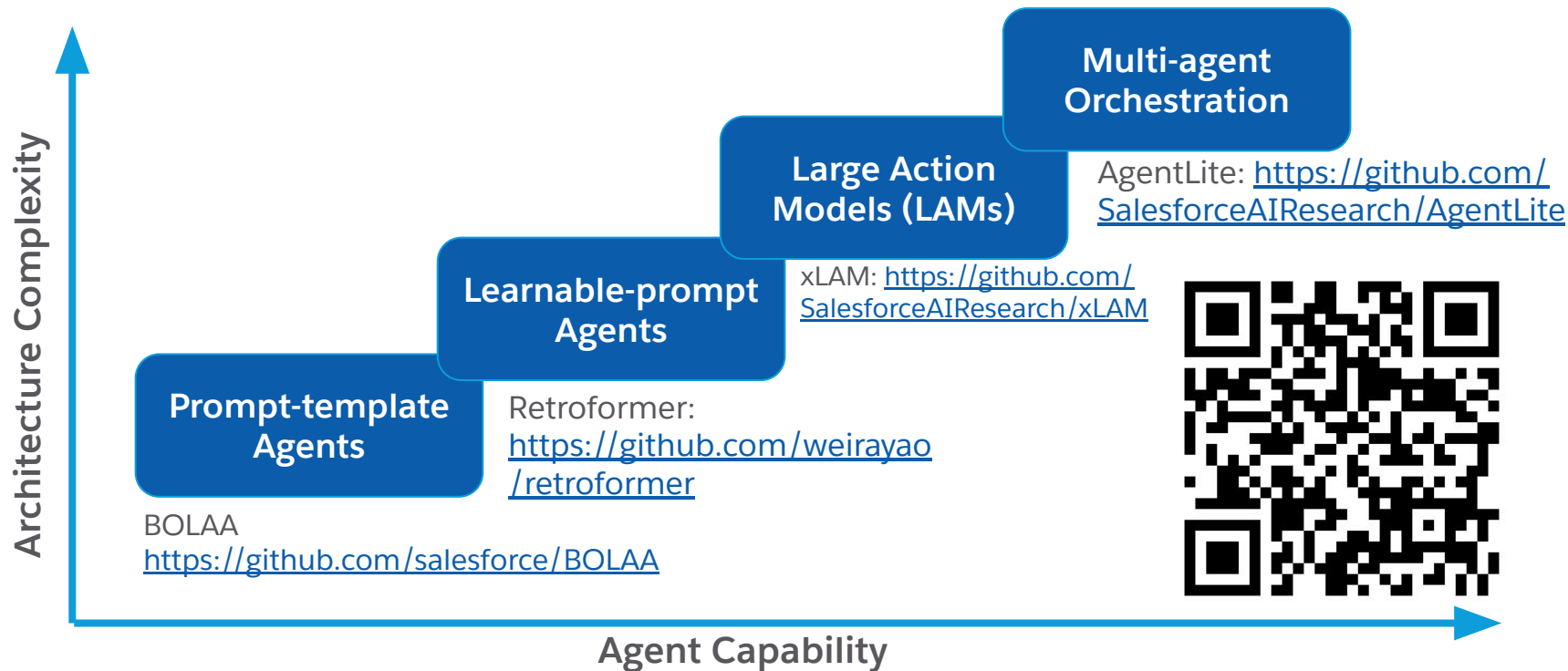
Resources & References

Selected papers & open source
implementations



Resources to Get Started

Salesforce AI Research Open Source Ecosystem



References



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Thank you

